ConvoBridge

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ConvoBridge

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Dedicated to my parents,

who have always been my biggest supporters

and a constant source of inspiration

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Praise to be ALLAH, the Cherisher and Lord of the World, Most gracious and Most Merciful

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Project Summary

ConvoBridge is an AI-powered video conferencing application designed to remove language barriers in online meetings. Its main goal is to help people who speak different languages communicate easily and clearly. The system includes a Speech-to-Text feature that converts spoken words into text instantly. It also provides real-time translation so participants can understand conversations in their preferred language. The Text-to-Speech feature converts translated text into natural-sounding audio. A noise reduction module removes background noise to ensure clear sound quality. During meetings, live multilingual subtitles are displayed for better understanding. At the end of each meeting, the system automatically generates AI-based summaries, key highlights, and action points. A built-in chatbot assistant allows users to ask questions from the meeting transcript and get quick answers. The platform also includes core features such as video and audio calls, screen sharing, file sharing, and host controls. ConvoBridge aims to make online meetings smarter, clearer, and more inclusive for everyone.

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Introduction

Chapter Overview

This chapter introduced ConvoBridge, an AI-powered multilingual video conferencing platform designed to overcome language barriers, background noise, and lack of meeting automation. It outlined the system's core objectives, key deliverables, technical constraints, and relevance to course modules, establishing ConvoBridge as a practical and scalable communication solution..

1.1 Introduction

Virtual communication is playing an increasing role in education, business, health care and even international relations. With video conferencing tools, individuals can communicate across cities and borders without needing to travel. But language is a major barrier in online communication. Different languages spoken by participants can lead to communication problems. This can result in misunderstanding, confusion and inefficiency in meetings. To overcome this problem, we have come up with **ConvoBridge**, an AI-based video conference application to overcome language barriers and enhance communication efficiency. ConvoBridge aims to facilitate communication between people speaking different languages through the use of state-of-the-art Artificial Intelligence technologies. ConvoBridge uses Speech-to-Text to transcribe speech into text. Real-time translation technology is then used to convert this text into the language of choice of the other users. To enhance accessibility, the translated text is also spoken out loud using Text-to-Speech technology. This allows users to both see and hear the translation. Furthermore, ConvoBridge offers real-time, multi-language subtitles during meetings, enabling users to keep up with the conversation. The system also has a noise reduction module to reduce background noise. The system also automatically creates an AI-generated summary of the discussion and action items after the meeting, so attendees can easily catch up on the key points. It also offers basic video conferencing features like video and audiocalls, screen sharing, file sharing and control options for the host. In addition, there is an AI chatbot to ask questions about the meeting transcript. In sum, ConvoBridge seeks to make video meetings smarter, more accessible and more inclusive, by leveraging communication technologies with Artificial Intelligence.

1.2 Significance of the Proposed System

ConvoBridge makes online communication faster, clearer, and more inclusive for everyone involved. For participants, it reduces confusion by providing real-time translation, live subtitles, and clear audio during meetings. Users can speak in their own language and understand others without difficulty. This creates a comfortable and stress-free communication environment. For professionals such as teachers, students, business teams, and international clients, ConvoBridge improves meeting efficiency. It automatically converts speech into text, translates it instantly, and generates meeting summaries with key points and action items. This helps users focus on discussion instead of worrying about language differences or taking detailed notes. For organizations and administrators, the system improves workflow and collaboration. Meetings become more structured with organized transcripts, summaries, and searchable records. The platform is designed to be scalable, meaning it can support small teams as well as large organizations in the future. Its simple deployment and manageable structure make it practical for real-world implementation.

1.2.1 For For individual users

For individual users, the system significantly reduces the frustration caused by language barriers and poor audio quality. Real-time translation, live multilingual subtitles, and noise reduction ensure that every participant can follow and contribute to discussions clearly and confidently, without requiring any technical expertise. Users can speak in their own language and instantly receive translated audio and text, creating a comfortable and stressfree communication experience.

1.2.2 For Professionals and Academic Communities

For professionals such as teachers, students, business teams, and international clients, ConvoBridge improves meeting efficiency by automating note-taking through AI-generated summaries, key highlights, and action items. This allows participants to remain fully focused on the discussion rather than manually recording information, significantly reducing the risk of missed decisions or overlooked follow-up tasks. The built-in chatbot assistant further allows users to quickly retrieve specific information from the meeting transcript at any time.

ConvoBridge Is an AI-powered multilingual video conferencing application

That provides real-time speech-to-text transcription, multilingual translation, noise reduction, text-to-speech audio output, live multilingual subtitles, automated AI-generated meeting summaries with key highlights and action points, and a built-in chatbot assistant, ensuring that every participant can speak, listen, and understand in their preferred language without any communication barriers, making virtual collaboration smarter, more productive, and fully inclusive **Unlike** existing platforms such as Zoom, Google Meet, and Microsoft Teams, which offer only limited or manually configured language support, lack intelligent noise reduction, provide no automated meeting summarization, and require third-party plugins for basic translation features,

Our product modules directly into a single, unified conferencing platform, delivering a seamless, noise-free, and multilingual communication experience without the need for any external tools or plugins, making it the most intelligent and inclusive video conferencing solution available for global users across education, healthcare, business, and government sectors. In a world where remote work, online education, and global collaboration have become the norm, the ability to communicate across language boundaries is no longer optional but essential. ConvoBridge directly contributes to reducing communication inequality by ensuring that non-native speakers, people with hearing difficulties, and participants from developing regions can engage equally and effectively in virtual meetings. Its inclusive design and intelligent automation make it a meaningful step toward bridging the global

1.3 Problem Statement

Today, many people use video conferencing platforms like Zoom, Google Meet, and Microsoft Teams to attend online meetings, classes, and business discussions. However, these platforms have many problems that make communication difficult for users around the world. The biggest problem is the language barrier. When people who speak different languages join the same meeting, they cannot understand each other clearly. This leads to confusion and miscommunication during important discussions. Another common problem is background noise. When a participant is in a noisy environment, the sound quality becomes very poor and other participants cannot hear them properly. Many platforms do not have a built-in system to automatically remove this background noise. Network issues are also a serious problem. When the internet connection is weak or suddenly drops, the meeting gets interrupted and participants miss important parts of the conversation. There is no automatic system in current platforms to reconnect and recover the lost content. Furthermore, most platforms do not provide any automatic meeting summary. After the meeting ends, participants have to remember or manually write down all the important points and decisions, which is very difficult and time-consuming. We are developing ConvoBridge to solve all these problems. ConvoBridge will provide real-time translation so that people speaking different languages can understand each other easily. It will also remove background noise to ensure clear audio quality during meetings. An auto-reconnect feature will handle network drops and restore the meeting automatically. At the end of each meeting, ConvoBridge will automatically generate a summary with key points and action items so that no important information is lost.

1.4 Scope and Objectives

The objectives of the proposed system are outlined as follows:

1.4.1 Real-Time Speech-to-Text (STT) Module

Develop a real-time Speech-to-Text (STT) module using the Whisper model that accurately transcribes spoken dialogue into digital text during live meeting sessions. The module shall support multiple languages and accents, maintain a Word Error Rate (WER) of 15% or below under standard audio conditions, and operate with a transcription latency of no more than two seconds per spoken utterance. The transcribed text shall be continuously displayed to all participants as live captions and stored in the meeting transcript database for post-meeting use.

1.4.2 Noise Reduction Pipeline

Integrate a noise reduction pipeline using RNNoise and WebRTC protocols that automatically detects and suppresses background noise, echo, and microphone interference in real time before audio is processed by the STT module. The noise reduction module shall achieve a minimum Signal-to-Noise Ratio (SNR) improvement of 80% compared to unprocessed audio, ensuring that speech clarity is maintained even in noisy environments such as public spaces, shared offices, or home settings.

1.4.3 Real-Time Multilingual Translation

Implement a real-time multilingual translation module that translates the transcribed text from the speaker's source language into each participant's individually selected preferred language. The system shall support a minimum of five widely spoken languages in its initial release, including English, Urdu, Arabic, French, and Chinese, with a translation response latency of under three seconds per utterance. Translated text shall be displayed simultaneously as on-screen subtitles and delivered as natural audio output through the Text-to-Speech module.

1.4.4 Text-to-Speech (TTS) Module

Develop a Text-to-Speech (TTS) module using Coqui TTS that converts translated text into natural, human-like audio output in the participant's preferred language. The module shall produce speech that is clearly intelligible, appropriately paced, and distinguishable from the original speaker's audio stream, with a synthesis latency of under two seconds.

1.4.5 AI-Based Summarization System

Build an AI-based summarization system powered by a GPT-based language model that automatically processes the complete meeting transcript at the end of each session and generates a structured post-meeting report, including summary, key points, and action items. The report shall be generated within sixty seconds and made available immediately.

1.4.6 Intelligent Chatbot Assistant

Develop an intelligent Chatbot Assistant that allows participants to query the meeting transcript using natural language. The chatbot shall retrieve accurate answers with at least 85% accuracy and respond within three seconds through a dedicated chat panel.

1.4.7 Cross-Platform Mobile Application

Design and deliver a cross-platform mobile application for Android and iOS integrating all AI modules along with conferencing features such as video/audio calls, screen sharing, file sharing, subtitles, and host controls. The application shall include secure authentication and a user-friendly interface designed using Figma and implemented with Flutter or React Native.

1.4.8 Auto Reconnect Mechanism

Implement an Auto Reconnect mechanism that restores connectivity within ten seconds after a network drop without losing transcript data. The system shall maintain session continuity and notify users of disconnection and reconnection status.

1.5 System Limitations / Constraints

Despite its comprehensive design, ConvoBridge is subject to several technical, operational, and resource-based constraints that define the boundaries of the system in its initial development phase.

1.5.1 Computational and Latency Constraints

The simultaneous execution of multiple AI modules — Speech-to-Text (Whisper), Noise Reduction (RNNoise), Real-Time Translation, Text-to-Speech (Coqui TTS), and AI Summarization — introduces significant computational overhead. Running these pipelines in real time requires high-performance backend infrastructure. On low-end devices or constrained network conditions, there may be perceptible delays in transcription, translation output, or audio synthesis, which could affect the user experience.

1.5.2 Language Coverage Limitations

While ConvoBridge targets multilingual communication, the accuracy and fluency of real-time translation is dependent on the underlying neural machine translation model. Languages with limited training data — particularly regional dialects, minority languages, and low-resource languages such as Pashto or Sindhi — may yield lower translation quality. The system does not claim to deliver equal performance across all supported language pairs.

1.5.3 Speech Recognition Accuracy

The Whisper-based Speech-to-Text module performs well in clean audio conditions; however, accuracy degrades in the presence of strong accents, overlapping speech (crosstalk), or audio environments where residual noise persists after RNNoise processing. The system cannot guarantee perfect transcription under all real-world conditions.

1.5.4 Network Dependency

Although ConvoBridge includes an Auto Reconnect mechanism to handle network drops, the system fundamentally depends on a stable internet connection for real-time AI module operation. The Auto Reconnect feature mitigates interruptions but does not eliminate packet loss, jitter, or latency introduced during reconnection. In regions with consistently unreliable connectivity, the experience may be degraded.

1.5.5 Summarization and Chatbot Scope

The AI summarization module and chatbot assistant operate exclusively on the meeting transcript generated within the session. They have no access to external knowledge sources or prior meeting history. The quality of summaries and chatbot responses is therefore directly bounded by the completeness and accuracy of the STT transcription.

1.5.6 Platform and Device Constraints

In its initial release, ConvoBridge is targeted primarily as a mobile-first application. Full web browser support, desktop optimization, and cross-platform parity are considered future development goals and are not guaranteed in the first version of the system.

1.5.7 Privacy and Data Security

As meeting transcripts, summaries, and user data are stored in cloud-based databases (Firebase or MongoDB), the system is subject to the data privacy policies and security protocols of the chosen cloud provider (Google Cloud or AWS). End-to-end encryption for meeting data and compliance with regional data protection regulations (e.g., GDPR) is planned but not fully implemented in the initial prototype.

1.5.8 Scalability in Early Phases

The system is designed with scalability as a long-term goal; however, during the early development and testing phases, the backend infrastructure may not support a large number of concurrent users without performance degradation. Load testing and optimization for high-concurrency scenarios are deferred to later project phases.

1.6 Project Deliverables

A fully functional cross-platform mobile application for Android and iOS that integrates all AI modules and core conferencing features including video/audio calls, screen sharing, file sharing, live multilingual subtitles, and host controls.

- A real time Speech to Text (STT) module powered by Whisper that accurately transcribes speech into text. spoken dialogue into digital text during live meeting sessions and displays live captions to all participants.

- A Noise Reduction module integrated with RNNoise and WebRTC that suppresses background noise, echo, and microphone interference in real time to ensure clear and disturbance-free audio throughout every meeting session.
- A Real-Time Translation module that translates transcribed speech into each participant's preferred language and delivers the output simultaneously as on-screen subtitles and natural audio through the Text-to-Speech module.
- A Text-to-Speech (TTS) module powered by Coqui TTS that synthesizes natural, human-like multilingual audio output from translated text, enabling smooth and uninterrupted multilingual communication between participants.
- An AI-based Meeting Summarization module that automatically generates a structured post-meeting report containing a concise summary, key highlights, and clearly identified action items at the end of each session without any manual input from participants.
- A Chatbot Assistant that allows participants to query the meeting transcript using natural language questions and receive accurate, context-relevant answers derived from the session transcript during or after the meeting.
- An Auto Reconnect mechanism that detects network drops during an active session and automatically restores connectivity without loss of transcript data or disruption to participant flow.
- A secure user authentication system supporting login, signup, and meeting access management through unique meeting IDs and shareable links.
- A cloud-deployed backend built with Python (Flask/FastAPI) and hosted on Google Cloud or AWS, providing scalable API endpoints that connect all AI modules with the mobile application frontend.
- A structured database system using Firebase or MongoDB for storing user profiles, meeting transcripts, and generated summaries securely in the cloud.

- A complete UI/UX design prototype created in Figma covering all application screens including login, meeting creation, active call view, subtitle display, and post-meeting summary report screens.
- A Software Requirements Specification (SRS) document detailing all functional and non-functional requirements of the ConvoBridge system.
- A final project report documenting the system architecture, module design, implementation details, testing results, and conclusions drawn from the development and evaluation of the ConvoBridge platform.

1.7 Relevance to Course Modules

This section outlines how ConvoBridge relates to the core course modules studied throughout the program. Each module contributes directly to the design, development, and deployment of the system.

1.7.1 Artificial intelligence

The core intelligence of ConvoBridge is built upon AI concepts studied in this course. The Speech-to-Text module using Whisper, the GPT-based summarization engine, and the chatbot assistant all rely on fundamental AI principles including machine learning, natural language processing, and neural network architectures. The study of AI algorithms and intelligent agent design directly informs the design and integration of these modules into the system.

1.7.2 Natural Language Processing

The real-time translation module, AI summarization engine, and chatbot assistant are direct applications of Natural Language Processing techniques. Concepts such as tokenization, sequence-to-sequence modeling, transformer architectures, and text summarization form the theoretical foundation of these components.

1.7.3 Software Engineering

The development lifecycle of ConvoBridge follows structured software engineering practices. Requirement analysis, system design, modular architecture, testing strategies, and deployment planning are applied throughout the project. The use of version control via GitHub, API design, and iterative development reflects modern software engineering principles.

1.7.4 Database Systems

ConvoBridge uses Firebase and MongoDB to store user profiles, meeting transcripts, and summaries. Database schema design, handling structured and unstructured data, and efficient data retrieval apply concepts of database design, NoSQL systems, and data modeling.

1.7.5 Computer Networks

The real-time communication features are built on WebRTC and Socket.IO, which rely on networking protocols. Concepts such as peer-to-peer communication, signaling, ICE, STUN, TURN, and TCP/UDP are applied in conferencing and auto reconnect features.

1.7.6 Human Computer Interaction

The mobile application interface is designed using HCI principles. UI/UX design with Figma, user-friendly screens, accessibility, and multilingual support reflect usability principles and user-centered design.

1.7.7 Operating Systems

The system handles concurrent execution of multiple modules such as STT, noise reduction, translation, and TTS. This applies operating system concepts like multi-threading, process scheduling, and inter-process communication.

1.7.8 Web and Mobile Application Development

The frontend is developed using Flutter or React Native, and the backend uses Python with Flask or FastAPI. Concepts such as RESTful APIs, client-server architecture, and responsive UI design are applied.

1.7.9 Information Security

ConvoBridge handles sensitive data such as transcripts and user profiles. Security principles like authentication, data encryption, and privacy protection are applied, with plans for end-to-end encryption.

1.7.10 Cloud Computing

The backend is deployed on cloud platforms such as Google Cloud or AWS. Concepts including cloud services, scalability, containerization, and cloud database management are applied.

1.8 Contribution

ConvoBridge is a novel AI-powered multilingual video conferencing platform that addresses the critical limitations of existing communication systems. The key contributions of this project are as follows:

1.8.1 Unified End-to-End AI Pipeline

ConvoBridge introduces a unified end-to-end AI pipeline that seamlessly integrates Speech-to-Text, Real-Time Translation, Text-to-Speech, Noise Reduction, and Summarization within a single platform, eliminating the need for external plugins or third-party tools.

1.8.2 Real-Time Multilingual Communication

The system enables real-time multilingual communication, allowing participants speaking different languages to interact simultaneously through live transcription, neural machine translation, and Text-to-Speech synthesis.

1.8.3 Intelligent Post-Meeting Reporting

An intelligent reporting mechanism is implemented to automatically generate structured meeting summaries, key highlights, and action items using a GPT-based summarization model, reducing the need for manual note-taking.

1.8.4 Context-Aware Chatbot Assistant

ConvoBridge includes a context-aware chatbot assistant that allows users to query meeting transcripts using natural language and instantly retrieve relevant information such as decisions, discussions, and assigned tasks.

1.8.5 Auto Reconnect Mechanism

The system incorporates a robust auto-reconnect mechanism that restores interrupted meeting sessions within seconds while maintaining transcript continuity and system stability.

1.8.6 Noise-Resilient Speech Processing

A noise-resilient speech processing pipeline is utilized, combining RNNoise and WebRTC-based noise suppression techniques to enhance audio clarity and improve transcription accuracy.

1.8.7 Cross-Platform Mobile Application

The platform is developed as a cross-platform, mobile-first application to ensure accessibility and usability for a wide range of users, including students, professionals, and multilingual communities.

1.9 Chapter Summary

This chapter introduces ConvoBridge, an AI-based video conferencing system designed to improve communication in online meetings. It highlights how language barriers, background noise, and lack of automation create confusion and reduce efficiency in existing platforms. To solve these issues, ConvoBridge uses advanced AI technologies such as Speech-to-Text, real-time translation, and Text-to-Speech to enable smooth multilingual communication. The system provides live subtitles, translated audio, and noise reduction to enhance clarity and understanding for users. It is especially beneficial for individuals, students, teachers, and professionals by making meetings more efficient and stress-free. Additionally, ConvoBridge automatically generates meeting summaries with key points and action items, reducing the need for manual note-taking. It also includes features like chatbot assistance, screen sharing, and file sharing for a complete meeting experience. The chapter further discusses the limitations of existing platforms and explains how ConvoBridge offers a better solution. Finally, it outlines the system's objectives, constraints, and deliverables, presenting it as a practical and scalable communication platform.

Chapter 2

Literature Review

Chapter Overview

This chapter reviewed existing literature relevant to ConvoBridge's core modules, covering foundational studies on speech transcription, noise suppression, real-time communication, multilingual translation, and meeting summarization. Model-based approaches and comparative analysis confirmed ConvoBridge as a unified and intelligent multilingual conferencing solution..

2.1 Foundational Studies on Speech and Communication Technologies

The problem of language barriers and communication inefficiency in video conferencing has been tackled by various researchers and developers through the application of Speech-to-Text, machine translation, noise reduction, and summarization technologies.

The challenge of accurate real-time speech transcription was addressed by Radford et al. who proposed Whisper, a large-scale weakly supervised model trained on 680,000 hours of multilingual and multitask speech data collected from the web. The model demonstrated near human-level robustness and accuracy across multiple languages and acoustic conditions, making it highly suitable for deployment in real-time transcription systems such as ConvoBridge [1].

The problem of background noise degrading speech quality in communication systems was addressed by Valin who proposed RNNoise, a hybrid approach combining traditional signal processing with a Recurrent Neural Network (RNN) to perform real-time noise suppression. The model operates at low computational cost and is capable of suppressing a wide variety of noise types including stationary and non-stationary noise, making it a practical solution for deployment in mobile conferencing applications [2].

The challenge of real-time peer-to-peer audio and video communication over the web was addressed by Bergkvist et al. who provided a comprehensive overview of WebRTC, a framework that enables direct browser-to-browser communication without external plugins. The authors described the core components including getUserMedia, RTCPeerConnection, and RTCDataChannel, as well as the underlying protocols ICE, STUN, and TURN that ensure reliable connectivity across diverse network environments [3].

The problem of multilingual neural machine translation and zero-shot translation performance was investigated by Zhang et al. who proposed architectural and training improvements for massively multilingual NMT models. Their work demonstrated significant gains in translation quality across low-resource language pairs, providing a strong theoretical foundation for the real-time translation module integrated into ConvoBridge [4].

The challenge of generating natural and intelligible speech synthesis across multiple languages was addressed by Casanova et al. who proposed YourTTS and Coqui TTS, open-source Text-to-Speech frameworks capable of producing high-quality multilingual speech output with low-latency inference. Their work demonstrated that open-source TTS systems can achieve near-commercial quality, making them viable for integration into multilingual conferencing platforms [5].

The problem of automatic meeting summarization and action item extraction from conversational transcripts was investigated by Feng et al. who proposed a transformer-based abstractive summarization model fine-tuned on meeting corpora. The model demonstrated strong performance in generating concise and coherent meeting summaries from raw transcripts, directly informing the design of the AI Summarization module in ConvoBridge [6].

The challenge of building intelligent question-answering chatbots over domain-specific documents was addressed by Devlin et al. who introduced BERT, a bidirectional transformer model pre-trained on large text corpora and fine-tuned for extractive question answering tasks. The architecture demonstrated state-of-the-art performance on reading comprehension benchmarks, providing the conceptual basis for the Chatbot Assistant module in ConvoBridge that retrieves answers directly from meeting transcripts [7].

2.2 Advanced Studies on Deployment and Multilingual Challenges

The problem of speaker diarization and multi-speaker transcription in multi-participant conferencing environments was investigated by Park et al. who proposed an end-to-end neural diarization system capable of identifying and separating overlapping speech from multiple speakers in real time. Their approach combined acoustic feature extraction with sequence-to-sequence modeling to assign speaker labels to transcribed segments, providing a theoretical foundation for handling multi-participant conversations in ConvoBridge where multiple speakers must be tracked simultaneously [8].

The challenge of low-latency streaming speech recognition in resource-constrained mobile environments was addressed by He et al. who proposed a streaming Conformer-based ASR architecture optimized for on-device deployment. The model achieved competitive word error rates while maintaining low memory footprint and inference latency, making it relevant to the mobile-first deployment strategy of ConvoBridge where real-time transcription must operate efficiently on mobile hardware [9].

The problem of cross-lingual subtitle generation and synchronization in live communication systems was examined by Jiang et al. who proposed a pipeline integrating ASR, neural machine translation, and subtitle rendering with temporal alignment. Their system demonstrated that end-to-end subtitle latency could be reduced significantly, directly informing the design of the live multilingual subtitle feature in ConvoBridge [10].

The challenge of automatic reconnection and session recovery in unstable network environments was studied by Apostolopoulos et al. who analyzed packet loss concealment and stream recovery mechanisms in real-time communication protocols. Their findings demonstrated that combining forward error correction with adaptive bitrate streaming significantly reduces perceptible disruption during network instability, providing a technical basis for the Auto Reconnect module in ConvoBridge

[11].

The problem of domain-specific vocabulary recognition in speech transcription systems was addressed by Peyser et al. who proposed a contextual biasing approach that injects domain-specific word lists into the ASR decoding process at inference time. This technique improved recognition accuracy for rare and technical terms without requiring full model retraining, which is particularly relevant for ConvoBridge deployments in specialized domains such as healthcare and legal conferencing [12].

The challenge of multilingual sentiment and intent detection within meeting transcripts was investigated by Conneau et al. who introduced XLM-R, a cross-lingual language model trained on 100 languages using masked language modeling. The model demonstrated strong zero-shot cross-lingual transfer on classification tasks, offering a pathway for ConvoBridge to extend its chatbot assistant with intent-aware responses across multiple languages [13].

The problem of real-time audio quality assessment and adaptive noise suppression under dynamic acoustic conditions was examined by Reddy et al. who organized the Deep Noise Suppression Challenge and proposed evaluation metrics for perceptual speech quality in noisy environments. Their benchmark established standardized criteria for noise suppression systems, providing a measurable evaluation framework applicable to the RNNoise-based noise reduction module integrated into ConvoBridge [14].

2.3 Model-Based Methods

A speech transcription approach [8] examined clean speech, stationary background noise, and non-stationary environmental noise conditions using an end-to-end deep learning pipeline, demonstrating robust accuracy and establishing a benchmark for real-time ASR in noisy conferencing environments.

A noise suppression approach [11] examined office, crowd, and mechanical noise environments using a hybrid RNN-based signal processing framework, demonstrating that recurrent architectures effectively separate speech from complex noise patterns in real-time mobile communication.

A multilingual translation approach [13] examined high-resource, low-resource, and zero-shot cross-lingual scenarios using a massively multilingual transformer, demonstrating significant improvements across diverse language pairs relevant to ConvoBridge's real-time translation module.

A meeting summarization approach [12] examined structured business meetings, informal discussions, and technical reviews using a transformer-based abstractive model fine-tuned on conversational corpora, producing coherent and concise summaries applicable to ConvoBridge's AI Summarization module.

A text-to-speech synthesis approach [14] examined single-speaker, multi-speaker, and cross-lingual voice cloning conditions using an end-to-end neural TTS framework, demonstrating near-commercial quality output suitable for ConvoBridge's multilingual audio generation feature.

Table 2.1 Comparison of Application Types, Weaknesses, and Proposed Solutions

Application Name	**Weakness**	**ConvoBridge**
Zoom	Relies on third-party plugins for translation; limited language support and no built-in summarization	Built-in real-time multilingual translation, AI-generated meeting summaries, and action item extraction without external plugins
Google Meet	Captions limited to very few languages; no speech translation or post-meeting intelligence	Full speech-to-text transcription with real-time translation and live multilingual subtitles
Microsoft Teams	Manual configuration required for translation; no integrated AI summarization or chatbot support	Automated translation pipeline with AI summarization, key highlights extraction, and a chatbot assistant for transcript queries
WhatsApp / Skype	No real-time multilingual support for live calls or chat	Real-time multilingual translation for both audio calls and chat messages
Existing STT Systems	Cannot handle background noise and unstable connectivity simultaneously; prone to transcript loss	RNNoise-based noise reduction with Auto Reconnect to preserve transcript continuity

2.4 Chapter Summary

This chapter reviewed literature relevant to ConvoBridge's core modules, covering speech transcription, noise suppression, real-time communication, multilingual translation, text-to-speech synthesis, meeting summarization, and question answering as the theoretical foundation. Further studies addressed specialized deployment challenges including speaker diarization, streaming ASR, subtitle generation, session recovery, contextual biasing, cross-lingual modeling, and noise evaluation benchmarks. Model-based approaches confirmed the viability of integrating deep learning for each core module. The comparative analysis in Table 2.1 highlighted existing platform limitations and justified ConvoBridge as a unified, intelligent, and multilingual conferencing solution.

Chapter 3

Requirement Analysis

Chapter Overview

This chapter presented the functional and non-functional requirements of ConvoBridge, covering user roles, meeting management, real-time transcription, multilingual translation, AI summarization, and external interface specifications, establishing a complete foundation for system development.

3.1 User Classes and Characteristics

ConvoBridge is designed to serve multiple types of users, each with different goals and levels of technical expertise.

- **Meeting Host:** The primary user who creates and manages meetings. The host controls participant access, mutes or removes participants, and manages screen and file sharing. This user is expected to have basic smartphone skills and a stable internet connection. The host also reviews the AI-generated meeting summary and action items after the session ends.
- **Meeting Participant:** A regular attendee who joins a meeting using a meeting ID or a shared link. This user speaks and listens during the session and may select a preferred language for translation and audio output. No technical expertise is required. The participant can also use the built-in chatbot to ask questions from the meeting transcript.
- **Multilingual User:** A participant or host who does not speak English as a first language. This user relies heavily on the real-time translation, live subtitles, and Text-to-Speech features to follow and contribute to the conversation. The interface must be simple and easy to navigate without language knowledge.
- **Guest User:** Joins a meeting without registering an account. This user enters only a display name to join. Guest users can attend calls but cannot create meetings or access post-meeting summaries. Their access is limited to the active session only.

- **System Administrator:** Manages user accounts, monitors platform performance, and handles backend configurations. This user has technical knowledge and accesses system settings and database records. Administrator functionality is planned for future development phases.

3.2 Requirement Identifying Technique

For ConvoBridge, two requirement identifying techniques were used: Use Case Analysis and Event-Response Table. These two techniques were selected because ConvoBridge is both an interactive end-user application and a real-time backend processing system.

Use Case Analysis: was the primary technique used because ConvoBridge is a user-facing mobile application where different types of users such as the host, participant, and guest user interact with the system in different ways. Use cases helped identify what each user needs to do, such as joining a meeting, selecting a preferred language, or querying the chatbot. A use case diagram was created to show these interactions visually, and detailed use case descriptions were written to capture the flow of each user action, the expected system response, and any alternative or error conditions.

Event-Response Table: was used as a secondary technique because many core features of ConvoBridge operate in the background in real time without direct user input. For example, when a participant speaks, the system automatically captures audio, reduces background noise, converts speech to text, translates it, and delivers the output as subtitles and audio. These backend processes are triggered by events rather than user clicks. An event-response table helped capture these automatic system behaviors clearly by listing each triggering event alongside the expected system response.

Together, these two techniques provided complete coverage of both the user-facing interactions and the backend real-time processing requirements of the ConvoBridge system.

3.3 Functional Requirements

Functional requirements describe what a system or software is supposed to do — the core features, actions, and behaviors it must perform to meet user and business needs. This section presents the core functional requirements of ConvoBridge in a concise and structured form.

3.3.1 User Registration

The system shall allow new users to create an account securely. This ensures that only valid and registered users can access the core features of ConvoBridge including meeting creation and post-meeting summaries.

Table 3.1 FR-1 – User Registration

Identifier	FR-1
Title	User Registration
Requirement	The new user shall be able to create an account by providing a valid name, email address, and password. The system shall validate the email format and ensure the password meets minimum length requirements before creating the account.
Source	Meeting Host, Meeting Participant
Rationale	Without registration, users cannot create meetings or access post-meeting summaries and transcripts.
Priority	High

Table 3.1 outlines FR-1, which specifies how the system handles new user registration. It describes the validation of email format and password strength before account creation, ensuring secure and reliable user onboarding.

3.3.2 User Login

The system shall allow registered users to securely log in using their credentials. The system enforces account lockout after multiple consecutive failed login attempts to prevent unauthorized access.

Table 3.2 FR-2 – User Login

Identifier	FR-2
Title	User Login
Requirement	The user shall log in using valid credentials. Invalid credentials will display an error message.
Source	Meeting Host, Meeting Participant
Rationale	Ensures secure and authenticated access to the system.
Priority	High

Table 3.2 outlines FR-2, which specifies how registered users securely access the system. It describes the lockout mechanism applied after multiple failed login attempts to prevent unauthorized access.

3.3.3 Guest Join Without Registration

The system shall allow external participants to join a meeting without creating an account. Guests can participate in audio and video but are restricted from accessing transcripts, summaries, or the chatbot assistant.

Table 3.3 FR-3 – Guest Join

Identifier	FR-3
Title	Guest Join
Requirement	Guest can join a meeting using a meeting ID and display name only.
Source	Guest User
Rationale	Improves accessibility for one-time participants.
Priority	Medium

Table 3.3 outlines FR-3, which allows external participants to join without an account. It describes the restricted access policy limiting guests to audio and video participation only.

3.3.4 Create Meeting

The system shall allow registered users to create a new meeting session with a unique meeting ID and shareable link. The host is automatically assigned host-level privileges upon meeting creation.

Table 3.4 FR-4 – Create Meeting

Identifier	FR-4
Title	Create Meeting
Requirement	Host can create a meeting with a unique ID and shareable link. Only registered users are allowed.
Source	Meeting Host
Rationale	Primary entry point for initiating communication on ConvoBridge.
Priority	High

Table 3.4 outlines FR-4, which enables registered users to initiate a new meeting session. It describes how the system generates a unique meeting ID and automatically assigns host-level privileges to the creator.

3.3.5 Join Meeting

The system shall allow both registered users and guests to join an active meeting using a meeting ID or shareable link. Users cannot join a meeting that has already ended.

Table 3.5 FR-5 – Join Meeting

Identifier	FR-5
Title	Join Meeting
Requirement	User joins via meeting ID or shareable link. Cannot join an ended meeting.
Source	All Users
Rationale	Provides flexible and easy access for all participants.
Priority	High

Table 3.5 outlines FR-5, which allows all users to enter an active meeting session. It describes the access control rule preventing participants from joining a meeting already terminated by the host.

3.3.6 Host Controls

The system shall provide the meeting host with administrative controls including muting participants, removing disruptive users, and ending the meeting at any time.

Table 3.6 FR-6 – Host Controls

Identifier	FR-6
Title	Host Controls
Requirement	Host can mute, remove participants, and end the meeting at any time. A removed user cannot rejoin.
Source	Meeting Host
Rationale	Maintains order and security during the session.
Priority	High

Table 3.6 outlines FR-6, which grants the host administrative controls over the session. It describes the host's exclusive rights to manage participants and terminate the meeting when necessary.

3.3.7 Real-Time Speech-to-Text Transcription

The system shall continuously convert spoken audio into text in real time, serving as the foundation for all AI-powered features including translation, summarization, and the chatbot assistant.

Table 3.7 FR-7 – Speech-to-Text Transcription

Identifier	FR-7
Title	STT Transcription
Requirement	System converts speech into text in real-time with a delay of $\leq 2\,\mathrm{s}$ and a Word Error Rate (WER) of $\leq 15\%$.
Source	All Participants
Rationale	Foundation for all AI-powered features in ConvoBridge.
Priority	High

Table 3.7 outlines FR-7, which specifies real-time conversion of spoken audio into text. It describes the latency and accuracy benchmarks the system must meet, and confirms that transcription activates automatically upon speech detection.

3.3.8 Noise Reduction

The system shall automatically filter and suppress background noise from audio input before STT processing. Noise reduction is always enabled and requires no manual configuration.

Table 3.8 FR-8 – Noise Reduction

Identifier	FR-8
Title	Noise Reduction
Requirement	System automatically removes background noise before STT processing. Always enabled with no manual configuration required.
Source	All Participants
Rationale	Improves transcription accuracy in noisy environments.
Priority	High

Table 3.8 outlines FR-8, which specifies automatic background noise suppression before audio reaches the STT engine. It describes how the module operates continuously without requiring any user activation.

3.3.9 Preferred Language Selection

The system shall allow each user to select their preferred language before or during a meeting, determining the target language for both real-time translation and text-to-speech output.

Table 3.9 FR-9 – Language Selection

Identifier	FR-9
Title	Language Selection
Requirement	User selects a preferred language for translation and TTS output. Minimum 5 languages supported.
Source	All Users
Rationale	Enables personalized multilingual communication for each participant.
Priority	High

Table 3.9 outlines FR-9, which allows each participant to select their preferred lan-

guage. It describes how this selection drives both the translation output and the text-to-speech audio delivery for that user.

3.3.10 Real-Time Translation

The system shall translate transcribed speech into each participant's preferred language within 3 seconds, removing language barriers in multilingual meetings.

Table 3.10 FR-10 – Real-Time Translation

Identifier	FR-10
Title	Real-Time Translation
Requirement	System translates speech into each user's selected language within 3 seconds.
Source	All Users
Rationale	Removes language barriers in multilingual meetings.
Priority	High

Table 3.10 outlines FR-10, which specifies real-time translation of transcribed speech. It describes how the system delivers personalized translation output to each participant simultaneously within the defined latency limit.

3.3.11 Text-to-Speech Audio Output

The system shall convert translated text into spoken audio and deliver it to the corresponding user within 2 seconds. TTS is only activated for users whose language differs from the speaker.

Table 3.11 FR-11 – Text-to-Speech Output

Identifier	FR-11
Title	TTS Output
Requirement	System converts translated text into audio within 2 seconds. Only activated for users in a different language session.
Source	All Users
Rationale	Assists users who cannot read on-screen subtitles during meetings.
Priority	High

Table 3.11 outlines FR-11, which specifies conversion of translated text into spoken audio. It describes the activation condition and latency requirement ensuring timely and language-appropriate audio delivery for each participant.

3.3.12 AI Meeting Summarization

The system shall automatically generate a structured summary upon meeting conclusion, including key discussion points and action items, eliminating the need for manual note-taking.

Table 3.12 FR-12 – AI Meeting Summarization

Identifier	FR-12
Title	Meeting Summary
Requirement	System generates a structured summary, key points, and action items within 60 seconds of meeting end using only the session transcript.
Source	All Users
Rationale	Eliminates the need for manual note-taking after meetings.
Priority	High

Table 3.12 outlines FR-12, which specifies automatic generation of a structured post-meeting report. It describes how the AI module processes the session transcript to

produce a concise summary and action items without any manual input.

3.3.13 Chatbot Assistant

The system shall provide an AI-powered chatbot that allows users to query the meeting transcript and receive relevant answers within 3 seconds, operating exclusively on the current session transcript.

Table 3.13 FR-13 – Chatbot Assistant

Identifier	FR-13
Title	Chatbot Assistant
Requirement	User can query the transcript and receive a context-relevant response within 3 seconds.
Source	Registered Users
Rationale	Enables quick information retrieval without interrupting the meeting flow.
Priority	Medium

Table 3.13 outlines FR-13, which provides an AI-powered chatbot for transcript queries. It describes how the chatbot retrieves accurate answers from the session transcript within the defined response time.

3.3.14 Screen and File Sharing

The system shall allow participants to share their screen or upload files during an active session. Only one screen share may be active at a time to avoid conflicts in the meeting view.

Table 3.14 FR-14 – Screen and File Sharing

Identifier	FR-14
Title	Screen/File Sharing
Requirement	Users can share their screen or upload files during an active meeting. Only one screen share active at a time.
Source	All Users
Rationale	Supports real-time visual collaboration without external tools.
Priority	Medium

Table 3.14 outlines FR-14, which enables screen and file sharing during meetings. It describes the single-share constraint applied to prevent view conflicts and ensure smooth collaborative work.

3.4 Non-Functional Requirements

Reliability:

- **R-1: System Stability** The system should operate reliably during video conferencing sessions without crashes, ensuring continuous communication between participants.
- **R-2: Auto-Reconnect Handling** In case of network failure, the system should automatically reconnect the user to the meeting without losing transcript or session data.

Usability:

- **U-1: User-Friendly Interface** The application should provide a simple and intuitive interface so that users can easily join meetings, select languages, and access features without confusion.
- **U-2: Real-Time Feedback** Features like subtitles, translation, and transcription should appear instantly to ensure a smooth and interactive user experience.

Performance:

- **P-1: Real-Time Processing** Speech-to-Text and translation modules should process input with minimal delay (within 2–3 seconds) for effective real-time communication.
- **P-2: Low Latency Communication** Video and audio streaming should maintain low latency to ensure smooth and uninterrupted conversations.

3.5 External Interface Requirements

This section describes how ConvoBridge communicates with users, hardware components, external software, and communication protocols.

3.5.1 User Interface Requirements

The ConvoBridge mobile application shall maintain a consistent white-themed interface with green and white as primary colors across all screens. Font styles, spacing, and contrast shall be clear and accessible for all users. The navigation flow shall follow a simple structure from authentication to meeting creation, active calls, transcripts, and summaries. Each screen shall use a clean, mobile-friendly layout with full-width buttons and easy navigation. The system shall provide clear and simple messages to guide users effectively. Live subtitles shall appear as a semi-transparent overlay at the bottom without blocking video content.

3.5.2 Software Interfaces

- **Frontend:** Flutter (Android & iOS cross-platform app).
- **Backend:** Firebase (Authentication, Cloud Functions, Firestore).
- **Database:** Cloud Firestore (stores user data, transcripts, summaries).
- **Authentication:** Firebase Authentication (Email/Password, OTP).
- **Notifications:** Firebase Cloud Messaging (real-time alerts).
- **AI Models:** Whisper (STT), RNNoise (noise reduction), Coqui TTS (TTS), GPT-based model (summarization/chatbot).

3.5.3 Hardware Interfaces

ConvoBridge runs on user devices such as Android/iOS smartphones and laptops with built-in microphone, speaker, and camera support for real-time video communication..

3.5.4 Communications Interfaces

- **HTTPS REST API:** All communication between the Flutter frontend and the Python backend shall use HTTPS for secure data transmission.
- **WebRTC:** Used with ICE, STUN, and TURN protocols for real-time peer-to-peer audio and video streaming between participants.
- **WebSocket via Socket.IO:** Used for real-time event signaling including meeting join, leave, mute, transcript updates, and subtitle delivery.
- **Email Notification:** The system shall send an email confirmation to the user upon successful account registration.

3.6 Chapter Summary

ConvoBridge supports different user types such as host, participant, multilingual users, guests, and future administrators with specific roles and permissions. The system provides secure registration and login, along with guest access for quick meeting entry. Hosts can create and manage meetings using unique IDs and have full control over participants. Key features include real-time speech-to-text transcription and automatic noise reduction to improve communication clarity. Overall, the system focuses on enabling smooth, secure, and multilingual real-time communication.

Chapter 4

Design and Architecture

Chapter Overview

This chapter presented the complete system design of ConvoBridge, covering architectural diagrams, data design, human interface screens, algorithms, tools and technologies, external APIs, and the cloud-based deployment environment used to implement the platform..

4.1 Modules

ConvoBridge is developed as a cross-platform mobile application using Flutter. The system is divided into functional modules based on core features of the application.

4.1.1 Mobile Application Modules (Flutter)

The ConvoBridge mobile application is developed using Flutter and consists of several functional modules to support real-time communication and AI-based features.

1: Authentication Module

- User registration using name, email, and password.
- Secure login with credential verification.
- Guest access using meeting ID and display name.
- Session management and logout functionality.

2:Meeting Management Module

- Create meetings with unique ID and shareable link.
- Join meetings via ID or link.
- Host controls (mute, remove participants, end meeting).
- Screen sharing and file sharing support.

3 :Real-Time Communication Module

- Audio and video communication using WebRTC.

- Live participant video streaming.
- Microphone and camera controls.
- Real-time signaling using Socket.IO.

4 :Audio Processing Module

- Capture live microphone audio.
- Apply noise reduction using RNNoise.
- Forward cleaned audio for transcription.

5 :AI Processing Module

- Speech-to-text conversion using Whisper model.
- Real-time translation of spoken text.
- Text-to-speech output using Coqui TTS.

6 :AI Services Module

- Generate meeting summaries after session ends.
- Extract key points and action items.
- AI chatbot for transcript-based queries.

7 :Notification Module

- Real-time push notifications using Firebase Cloud Messaging.
- Alerts for meeting updates and summaries.

4.1.2 Module 2: Backend Server Modules

The backend of ConvoBridge is developed using Firebase services and integrated third-party APIs such as Agora for real-time communication and pre-trained AI models for intelligent processing. The system is structured into modular components for scalability and efficiency.

Authentication and Authorization Module

- Handles user registration, login, and guest session access.
- Uses Firebase Authentication for secure login and token management.
- Ensures secure communication between Flutter app and backend services.

Meeting Management Module

- Creates and manages meetings with unique meeting IDs.
- Controls joining and leaving of participants in real time.
- Maintains meeting status and host permissions using Firebase Firestore.

Real-Time Communication Module

- Provides audio and video calling using Agora SDK.
- Manages real-time streaming between participants.
- Ensures low-latency and stable communication during meetings.

AI Processing Module

- Uses pre-trained Whisper model for speech-to-text conversion.
- Applies real-time translation using AI-based language models.
- Generates text-to-speech output using Coqui TTS model.
- Supports chatbot responses based on meeting transcripts.

Notification Service Module

- Sends push notifications using Firebase Cloud Messaging (FCM).
- Sends email notifications via Resend API.
- Notifies users about meeting updates, summaries, and alerts.

Analytics and Reporting Module

- Tracks meeting activity and user engagement using Firebase Firestore.
- Monitors system usage and AI feature performance.
- Provides data for future system improvements.

4.1.3 Module 3: Database Modules (Firebase Firestore)

1: User Data

- Stores user profiles (name, email, role).
- Manages authentication and session data.

2: Meeting Data

- Stores meeting ID, host, and participants.
- Tracks meeting status (active/ended).

3: Transcript Data

- Stores real-time speech-to-text data.
- Keeps full meeting conversation history.

4: AI Output Data

- Stores translations and summaries.
- Saves chatbot and AI-generated results.

5: Notification Data

- Stores push notification records.
- Tracks meeting and system alerts.

4.2 Architectural Design

The architecture of ConvoBridge follows a three-tier modular structure consisting of the Client Layer, Application Layer, and Data Layer. This design ensures scalability, real-time performance, and seamless integration of AI-powered communication features.

4.2.1 1. Client Layer (Mobile Application)

Technology: Flutter (Android & iOS)

The Client Layer provides the user interface for all types of users including host, participant, and guest. It enables real-time communication, meeting management, and AI-based interaction features.

Major Responsibilities:

- User authentication (login, registration, guest access)
- Meeting creation and joining via ID or link
- Live audio/video communication interface
- Display of real-time captions and translations
- Chatbot interaction with meeting transcript

Interfaces:

- Communicates with backend using REST APIs over HTTPS
- Real-time communication using WebRTC
- Event handling via Socket.IO
- Push notifications via Firebase Cloud Messaging (FCM)

4.2.2 2. Application Layer (Backend Services)

Technology: Firebase Functions / Node.js + AI APIs

The Application Layer handles business logic, real-time processing, and AI integration for ConvoBridge.

Key Functional Components:

- **Authentication & Security:** Firebase Authentication with token-based access control
- **Meeting Management:** Handles meeting lifecycle (create, join, end)
- **AI Processing Pipeline:**
  - RNNoise for noise reduction
  - Whisper for speech-to-text conversion
  - AI translation models for multilingual support
  - Coqui TTS for text-to-speech output
- **Chatbot Service:** Answers queries using meeting transcript

- **Notification Service:** Sends real-time alerts via FCM and email via Resend API

4.2.3 3. Real-Time Communication Layer

Technology: Agora SDK + WebRTC + Socket.IO

This layer manages real-time audio and video communication between participants.

Responsibilities:

- Peer-to-peer audio and video streaming using Agora/WebRTC
- Low-latency communication between users
- Real-time event signaling (join, leave, mute, subtitle updates)

4.2.4 4. Data Layer

Technology: Firebase Firestore

The Data Layer stores all persistent and session-based data for ConvoBridge.

Main Entities:

- Users (host, participant, guest profiles)
- Meetings (IDs, status, host info)
- Transcripts (real-time speech-to-text data)
- Summaries (AI-generated meeting summaries)
- Notifications (alerts and system messages)

4.2.5 5. External Services

- Firebase Authentication: User login and security
- Firebase Cloud Messaging (FCM): Push notifications
- Agora SDK: Real-time audio/video communication
- Resend API: Email notifications and verification
- AI Models: Whisper, RNNoise, Coqui TTS, GPT-based summarization

4.2.6 6. Data Flow Summary

- User logs in or joins as guest
- Meeting is created or joined via ID/link
- Audio/video is transmitted via Agora/WebRTC
- Audio is processed through AI pipeline (noise reduction → STT → translation → TTS)
- Transcript is stored in Firestore
- AI summary and chatbot responses are generated
- Notifications are sent to users

4.2.7 7. Architectural Style

- Client–Server based 3-tier architecture
- Real-time event-driven communication
- RESTful APIs over HTTPS
- Secure token-based authentication

4.2.8 8. Benefits of the Architecture

- Highly scalable and modular design
- Real-time multilingual communication support
- Secure and authenticated system
- Easy integration of AI models
- Independent scaling of each system layer

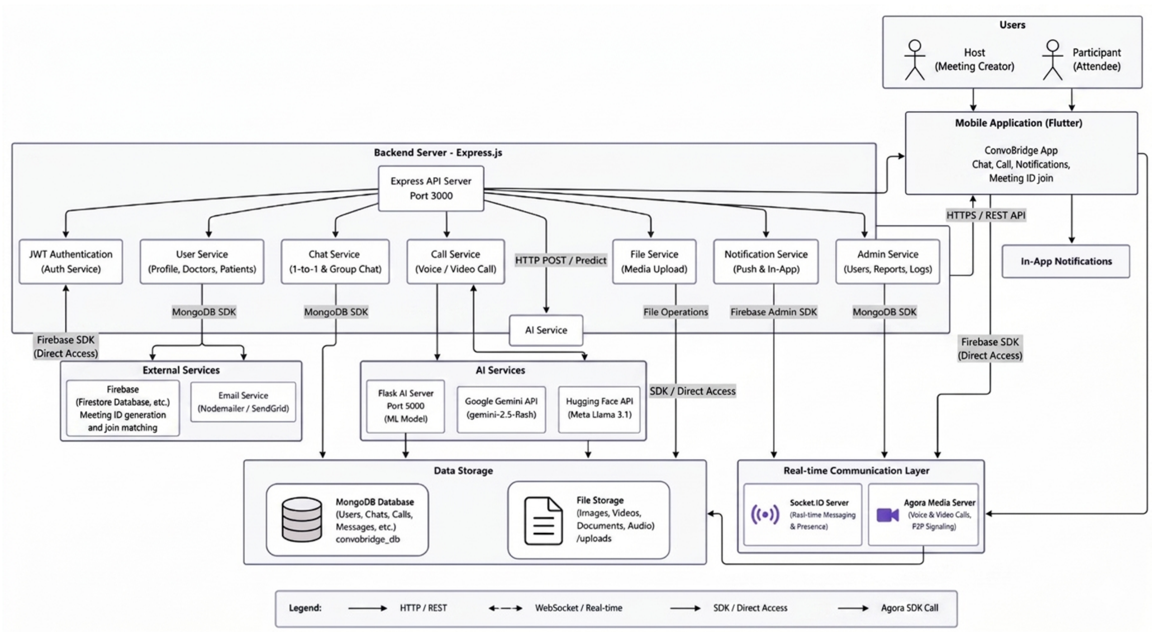

Figure 4.1 System Architecture diagram

4.3 Design Models

This section presents the design models developed for the ConvoBridge App, which help visualize system behavior, interactions, data flow, and architectural structure. These models ensure clarity, correctness, and consistency between requirements and implementation.

4.3.1 Activity Diagram

The Activity Diagram for ConvoBridge (see Figure 4.2) explains the complete workflow followed by users, including the Meeting Host, Participants, and Guest Users, as they interact with the system. It visually represents how the application handles authentication, meeting management, real-time AI processing, post-meeting summarization, and chatbot interaction.

The process begins when the Meeting Host creates a new meeting. The system then checks whether the user is a registered account holder or a guest. If the user is registered, they log in using valid credentials. If the user is a guest, they enter a display name to proceed. Both paths converge at the main decision point, where the system determines the next set of operations based on user interaction.

Meeting Flow: If the selected action is to join a meeting, participants enter the active session using a meeting ID or shared link. Once inside, when a speaker starts speaking, the system captures the audio and applies noise reduction using RNNoise to remove background interference. The cleaned audio is then passed to the Whisper Speech-to-Text module, which converts it into text in real time. The transcribed text is translated into each participant's preferred language using the Neural Machine Translation module. The system simultaneously delivers three outputs: multilingual subtitles displayed on screen, natural audio output generated by Coqui Text-to-Speech (TTS), and the transcript stored in the cloud database. This process continues in a loop until the host ends the session.

Post-Meeting Flow: When the meeting ends, the system processes the stored transcript using a GPT-based summarization model. The model generates a structured post-meeting report containing a concise summary, key discussion highlights, and clearly identified action items. This report is made available to the host and participants for in-app review and download.

Language Selection Flow: At any point during the meeting, a host or participant can select or change their preferred language. The system updates the translation and Text-to-Speech output accordingly for that individual participant without disrupting the ongoing session.

Host Controls Flow: The host can manage the meeting at any time by muting participants, removing participants, or ending the session. These controls are accessible directly from the active call screen.

Chatbot Flow: Users can access the Chatbot Assistant during or after the meeting. The user enters a natural language query through the chatbot panel. The system analyzes the query, retrieves relevant information from the meeting transcript, and delivers an accurate response within three seconds. The user can continue querying until all required information is obtained.

The activity ends when the user has reviewed the post-meeting summary or completed chatbot queries and exits the application, as illustrated in the activity diagram

(Figure 4.2).

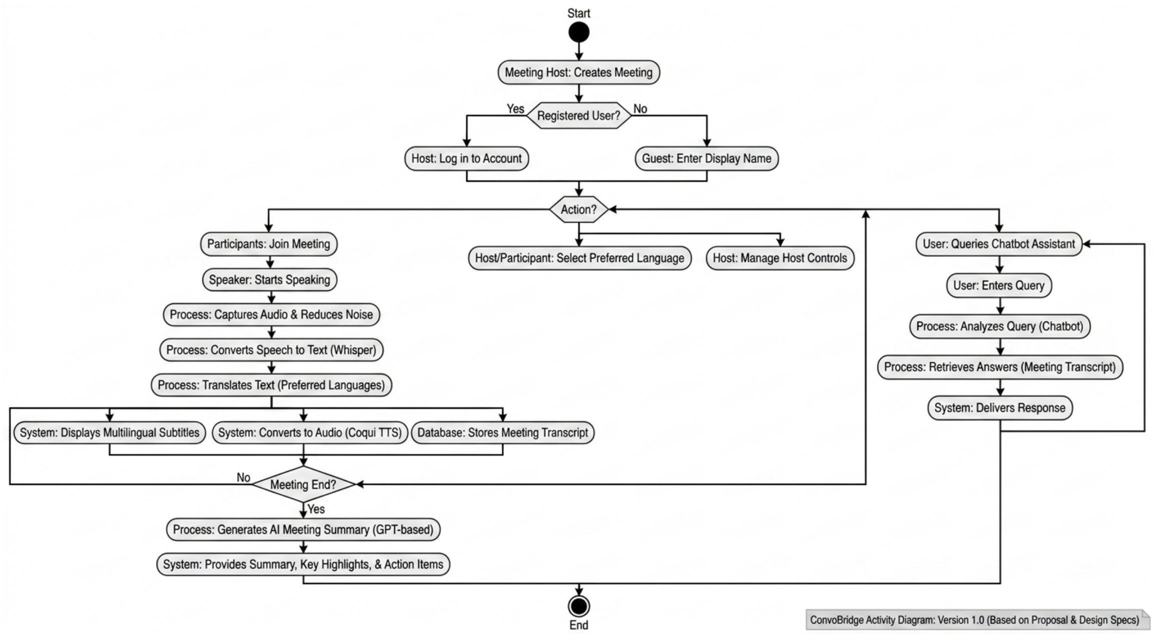

Figure 4.2 Activity Diagram of ConvoBridge App

4.3.2 Use Case Diagram

The Use Case Diagram for ConvoBridge (see Figure 4.3) presents a high-level representation of the interactions between system actors and core functionalities. The diagram identifies four primary actors: the Meeting Host, Meeting Participant, Multilingual User, and Guest User. Additionally, two secondary actors interact with the system: the Cloud Database and the API/Model Services, which include Whisper, RNNoise, Coqui Text-to-Speech (TTS), and the GPT-based model.

Meeting Host: The Meeting Host is the primary actor responsible for initiating system activities. The host can create a new meeting session, which generates a unique meeting ID and a shareable link. The host also manages the session through the *Manage Host Controls* use case, which includes muting participants, removing participants, and terminating the meeting at any time.

Meeting Participant: The Meeting Participant joins an active meeting using a meeting ID or shared link. Once connected, the participant engages in the central use case, *Participate in Multilingual Video Conference*, which encompasses multiple automated system processes executed during the session.

Multilingual User: A Multilingual User, who may act as either a host or a participant, interacts with the *Select Preferred Language* use case. This enables users to choose their desired language for translated subtitles and Text-to-Speech audio output in real time.

Guest User: The Guest User can join a meeting without account registration by providing a display name. However, guest users have restricted access and cannot utilize post-meeting intelligence features or the chatbot assistant.

Core System Use Cases (within Participate in Multilingual Video Conference):

The central use case includes multiple system-level processes connected through include relationships. These processes involve speech-to-text transcription using Whisper, noise reduction using RNNoise, multilingual translation, and generation of audio and subtitle outputs via Coqui TTS. Additionally, core conferencing functionalities such as video communication, screen sharing, and file sharing are provided. All generated transcripts are stored in the cloud database through the *Store Meeting Transcript* use case.

Post-Meeting Intelligence: After the meeting concludes, the Generate Meeting Summar use case is triggered. This process utilizes outputs from the speech-to-text, noise reduction, and translation modules, supported by the API/Model Services actor. The system produces a structured summary containing key discussion points, highlights, and action items. The Retrieve Post-Meeting Intelligence use case enables authorized users (host and participants) to access the stored summaries from the cloud database.

Chatbot Assistant: The Process Chatbot Query use case allows users to interact with the Chatbot Assistant by submitting natural language queries. The system processes these queries and retrieves relevant information directly from the stored meeting transcript, providing accurate and context-aware responses.

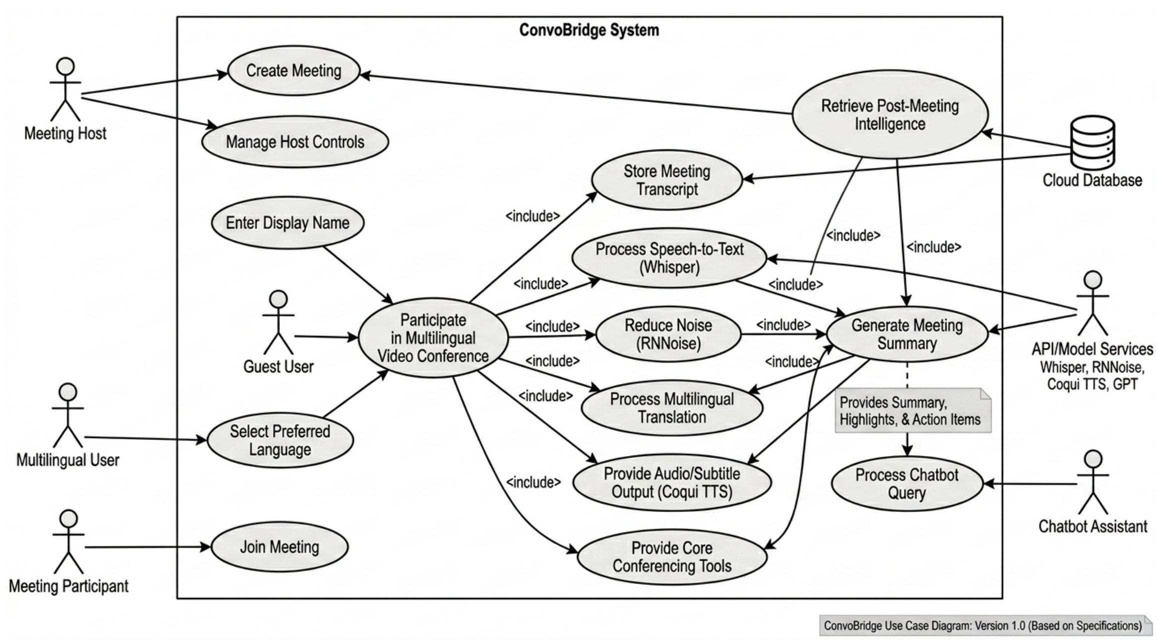

Figure 4.3 Use Case Diagram of ConvoBridge

4.3.3 Sequence Diagram

The Sequence Diagram for ConvoBridge (see Figure 4.4) illustrates the detailed chronological interaction between all system components during a complete meeting session. The diagram involves seven key components: the User (Speaker), the Meeting Interface built in Flutter, the AIEngineWrapper which is the central Python engine, the STT/Noise Engine powered by Whisper and RNNoise, the Translation/TTS Engine powered by Google and Coqui, the Summarization and Chatbot Module powered by GPT, the DatabaseOperationsHandler, and the Cloud Database for storing transcripts and profiles.

Real-Time Audio Processing Flow: The sequence begins when the user starts speaking and sends an audio stream to the Meeting Interface. The Flutter app forwards the audio asynchronously to the AIEngineWrapper which initiates a parallel AI pipeline. The STT/Noise Engine receives the audio, applies noise reduction using RNNoise, and transcribes it using Whisper. Once a transcript fragment is ready, it is passed back to the AIEngineWrapper asynchronously. The core audio and video stream is simultaneously forwarded to the Translation/TTS Engine.

Translation and Subtitle Delivery Flow: The AIEngineWrapper enters a loop for fragment processing where each transcribed fragment is sent to the Translation/TTS Engine for translation into the target language. The translated text is returned and the system updates the multilingual subtitles on the Meeting Interface. The transcript fragment is also stored in the DatabaseOperationsHandler. Subtitle metadata is added to the audio/video stream and multilingual subtitles along with multilingual audio output through Coqui TTS are delivered back to the user.

Post-Meeting Summarization Flow: When the user ends the call or stops speaking, the Meeting Interface triggers the summarization process through the AIEngineWrapper. The AIEngineWrapper fetches the complete meeting transcript from the DatabaseOperationsHandler which retrieves it from the Cloud Database. The full transcript is then sent to the Summarization and Chatbot Module which generates an AI-based meeting summary. The post-meeting intelligence including the summary, highlights, and action items is returned to the AIEngineWrapper. In an asynchronous post-processing step, the summary is stored back into the DatabaseOperationsHandler and persisted in the Cloud Database.

Summary Access Flow: After the meeting ends, the user accesses the post-meeting summary through the Meeting Interface. The interface fetches the PostMeetingIntelligence from the AIEngineWrapper which retrieves the stored summary and delivers it back to the user. The user can then view the complete meeting summary, key highlights, and action items directly within the application.

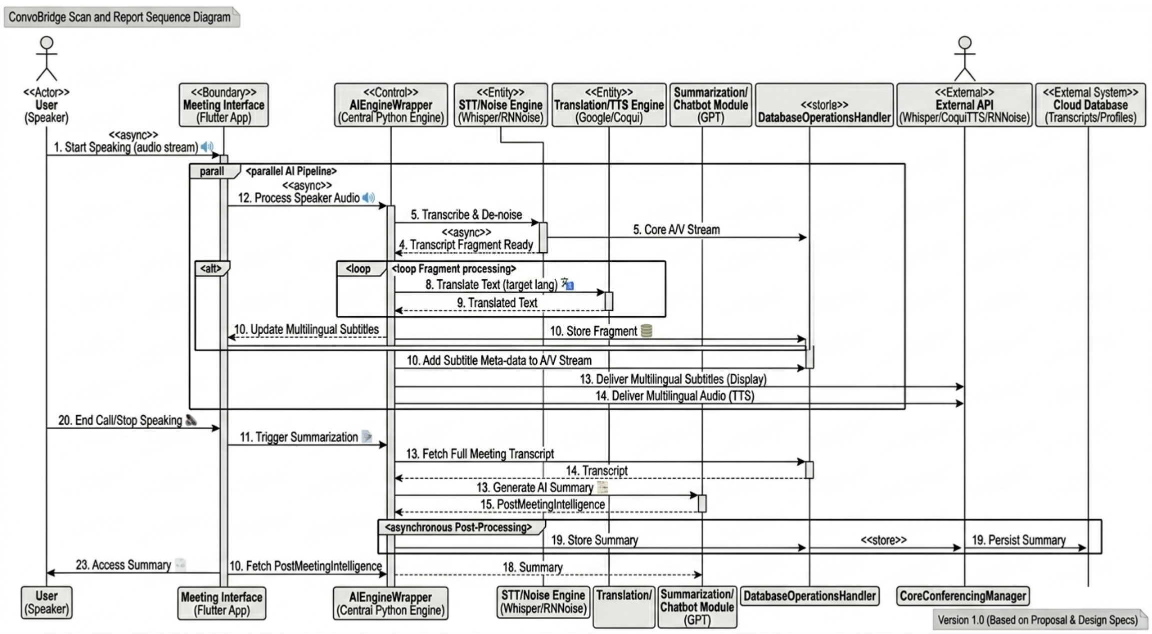

Figure 4.4 Sequence Diagram of ConvoBridge

4.3.4 State Transition Diagram

The State Transition Diagram for ConvoBridge (see Figure 4.5) illustrates all the possible states of the system and the transitions between those states based on user actions and system events. The diagram covers the complete lifecycle of the application from launch to session termination.

Authentication State: The system enters the Authentication state when the application is launched. If the user is new, they transition from the Login Screen to the Register state. After registration, the system moves to OTP Verification where the entered credentials are verified. Upon successful verification, the system transitions to the Authenticated state. If verification fails, the system returns to the Login Screen for re-entry of credentials.

Home Dashboard State: Once authenticated, the system enters the Home Dashboard and reaches an Idle state. From the Idle state, the user can trigger five different modules depending on their intended action. Each module runs independently and returns the system to the Idle state upon completion.

Meeting Module: When the user starts a call, the system transitions into the Meeting Module. The states within this module follow the sequence: Calling, then Ringing, then Connected upon accepting the call. If the call is rejected, the system returns to Ringing. From the Connected state the user can access all ConvoBridge AI features including real-time transcription, translation, subtitles, and TTS audio. When the host ends the session, the system transitions to the End Call state and finally to the Call Ended state before returning to Idle.

File Module: When the user selects Upload or Download File from the dashboard, the system enters the File Module. The states progress through Select File and Uploading. If the upload is successful, the system reaches the File Uploaded state. If an error occurs, it transitions to the Upload Failed state. The user can also transition to the Download File state. All paths return to Idle after completion.

Notification Module: When the user selects View Notifications, the system enters the Notification Module. The states progress through Receive Notification, View

Notification, and Mark as Read before returning to the Idle state.

Profile Module: When the user selects Manage Profile, the system enters the Profile Module. The states progress through View Profile, Edit Profile, and Save Changes before returning to the Idle state.

Logout and Session End: When the user selects Logout from the Home Dashboard, the system transitions to the Logging Out state. Once the session token is invalidated, the system moves to the Session Ended state which is the final state of the application lifecycle.

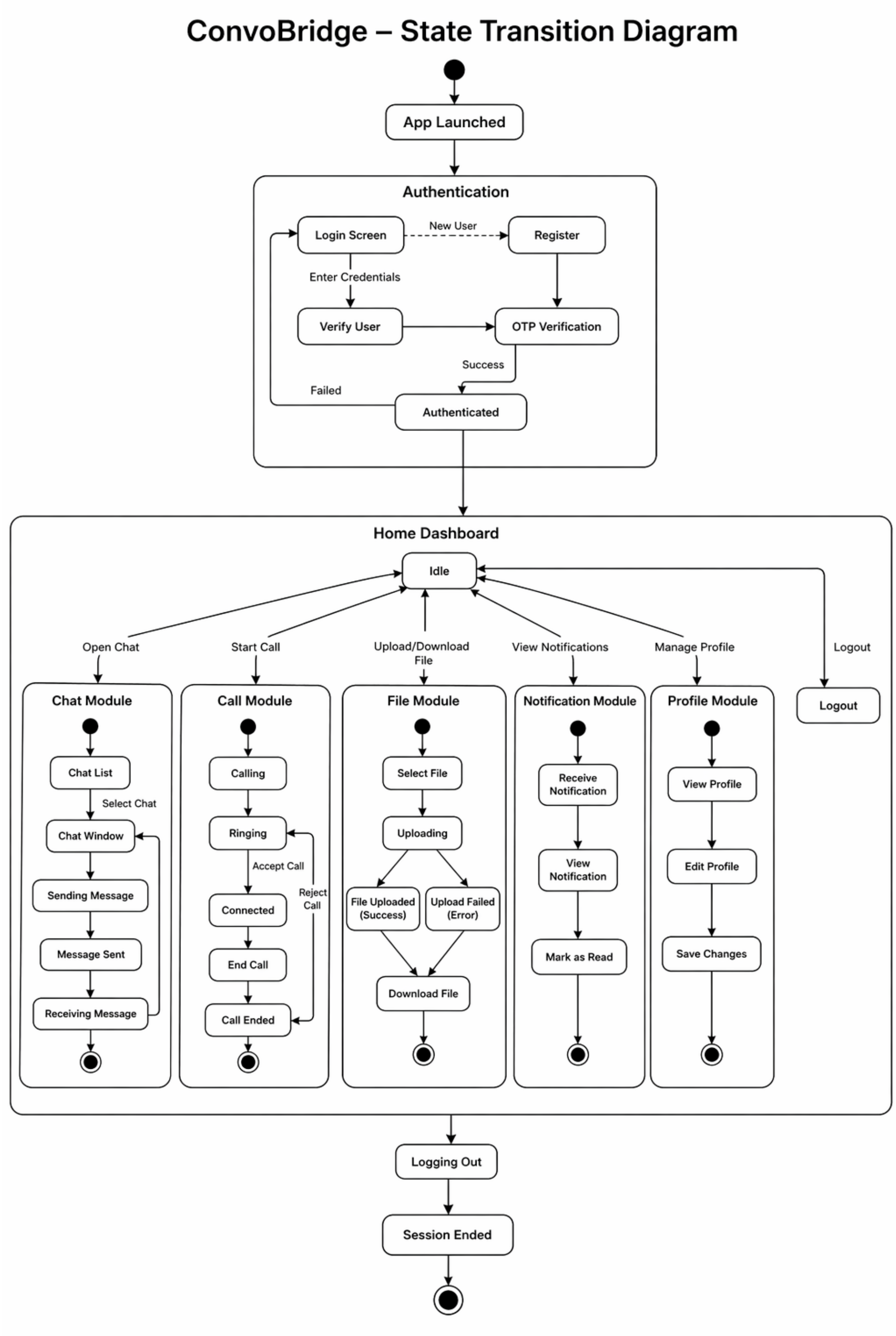

Figure 4.5 State Transition Diagram of ConvoBridge

4.4 Data Design

The data design of ConvoBridge transforms the system's information domain into well-structured data entities that support real-time communication, AI processing, multilingual translation, meeting summarization, and chatbot interaction. The backend uses a cloud-based NoSQL database such as Firebase or MongoDB to store user accounts, meeting transcripts, AI processing results, post-meeting summaries, and authentication data.

All system data is organized into the following major components:

- **User Data:** Stores information about meeting hosts, participants, and guest users.
- **Meeting Data:** Stores meeting session metadata including meeting ID, host ID, and participant list.
- **Transcript Data:** Stores real-time transcribed text fragments generated during active sessions.
- **Summary Data:** Stores AI-generated post-meeting summaries, key highlights, and action items.
- **Chatbot Query Data:** Stores user queries and retrieved answers from the meeting transcript.
- **Authentication Data:** Stores email, hashed passwords, and session tokens.

4.4.1 Data Dictionary

Table 4.1 Data Dictionary for ConvoBridge

S.No	**Entity / Attribute**	**Type**	**Description**
1	User	Object	Base entity representing any registered user in the system.
2	userId	String	Unique identifier for each user.
3	name	String	Full name of the user.

S.No	Entity / Attribute	Type	Description
4	email	String	Registered email address of the user.
5	preferredLanguage	String	Language selected for translation and TTS output.
6	role	String	Role of the user: Host, Participant, or Guest.
7	Authentication	Object	Contains login and registration details for each user.
8	passwordHash	String	Encrypted version of the user password.
9	sessionToken	String	Token generated upon successful login.
10	tokenExpiry	DateTime	Timestamp when the session token expires.
11	Meeting	Object	Represents an active or completed meeting session.
12	meetingId	String	Unique identifier for each meeting session.
13	hostId	String	userId of the meeting host.
14	status	String	Current status of the meeting: Active or Ended.
15	meetingLink	String	Shareable link for joining the meeting.
16	MeetingTranscript	Object	Stores the complete transcribed text of a session.
17	transcriptId	String	Unique identifier for each transcript.
18	transcriptText	Text	Full transcribed text generated during the meeting.
19	language	String	Source language of the transcribed content.
20	PostMeetingSummary	Object	Stores the AI-generated post-meeting report.
21	summaryId	String	Unique identifier for each summary.
22	summaryText	Text	Concise meeting summary generated by GPT model.
23	keyHighlightsList	List<String>	Key discussion points extracted from the transcript.

S.No	Entity / Attribute	Type	Description
24	actionItemsList	List<String>	Action items identified from the meeting transcript.
25	generatedAt	DateTime	Timestamp when the summary was generated.
26	ChatbotQuery	Object	Stores user queries and chatbot responses.
27	queryId	String	Unique identifier for each chatbot query.
28	queryText	String	Natural language question entered by the user.
29	responseText	String	Answer retrieved from the transcript by the chatbot.
30	GuestUser	Object	User who joins without a registered account.
31	displayName	String	Name entered by the guest to identify themselves.
32	Notification	Object	Stores system notifications delivered to users.
33	notificationId	String	Unique identifier for each notification.
34	message	String	Content of the notification message.
35	isRead	Boolean	Flag indicating whether the notification has been read.

4.5 Human Interface Design

The Human Interface Design of ConvoBridge focuses on delivering a simple, intuitive, and inclusive experience for all user types including Meeting Hosts, Meeting Participants, Multilingual Users, and Guest Users. The interface is designed to support easy navigation, fast meeting workflows, clear instructions, and responsive feedback. All screens follow a consistent dark theme with blue and white as primary colors, clear typography, and accessible font sizes. The application supports accessibility features such as large icons, readable subtitle text, and high-contrast UI elements.

4.5.1 Functionality from the User's Perspective

The system provides three distinct user experiences based on user role.

Meeting Host Perspective:

From the host's perspective, the application allows the user to register and log in securely, create a new meeting session with a unique ID and shareable link, manage participants by muting or removing them, share screen and files during the session, and access the post-meeting summary after the session ends. Upon login, the host is taken to the Home Dashboard where the Create Meeting button generates a unique meeting ID and shareable link instantly displayed on screen with a copy option. During the active session, the host sees participant video feeds, live multilingual subtitles, and a control bar with full host controls. When the meeting ends, the host is automatically redirected to the Post-Meeting Summary Screen where the AI-generated summary, key highlights, and action items are displayed with a download option.

Meeting Participant Perspective:

From the participant's perspective, the application allows the user to join a meeting using a meeting ID or shared link, select a preferred language for translation and audio output, view live multilingual subtitles during the session, listen to Text-to-Speech audio in their preferred language, and use the chatbot to query the meeting transcript. Upon joining, the participant selects their preferred language from a drop-down list accessible within two taps. During the session, translated subtitles appear at the bottom of the screen in real time. The chatbot panel is accessible from the active call screen where the participant types a natural language question and receives an answer from the transcript within three seconds without disrupting the ongoing meeting.

Guest User Perspective: From the guest user's perspective, the application allows the user to join an active meeting without creating an account by entering a valid meeting ID and a display name. The guest can view live video feeds, read multilingual subtitles, and listen to TTS audio during the session. Guest users do not have access to post-meeting summaries, transcripts, or the chatbot assistant. Upon attempting to access these features, the system displays a clear message informing the guest that these features require a registered account.

4.5.2 Screen Images

This section displays the user interface screenshots of the system from the user's perspective. All screenshots are shown below for clarity

4.5.3 User Interface

The following figures present the actual implemented screens of the ConvoBridge mobile application developed in Flutter.

Figure 4.6 Splash Screen

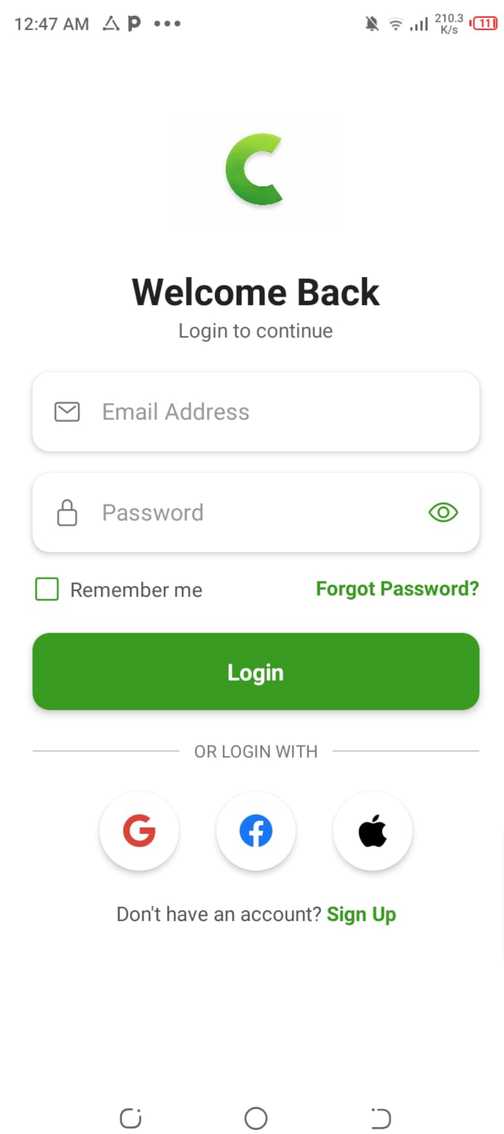

Figure 4.7 Login/Signup Screen

Figure 4.6shows the Splash Screen displayed on app launch, featuring the Convo-

Bridge logo with the tagline ”Breaking Language Barriers” and a countdown timer indicating loading progress..

Figure 4.7 shows the Login Screen where users enter their email and password to access the platform, with a ”Forgot Password?” link and third-party sign-in options via Google, Facebook, and Apple..

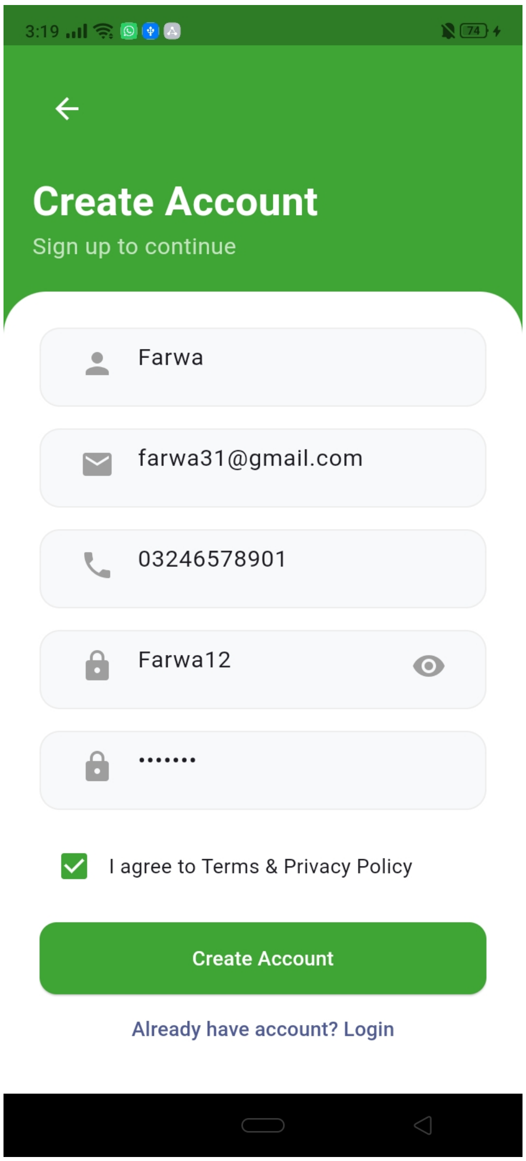

Figure 4.8 Create Account Screen

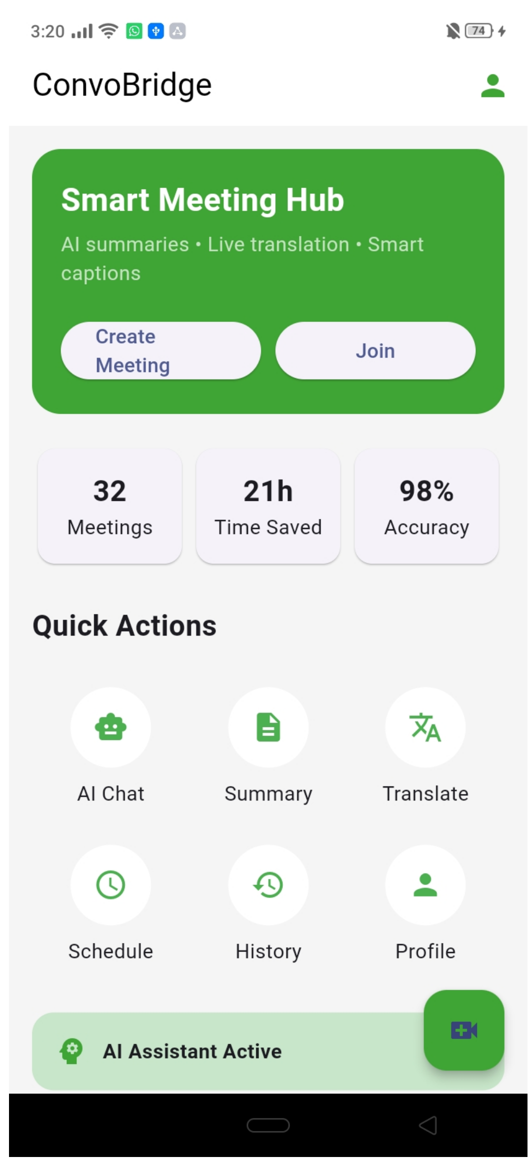

Figure 4.9 Home Dashboard

Figure 4.8 Figure 4.8 shows the Create Account Screen where new users register by enteringtheir name, email address, phone number, password, and a confirmation password in the provided fields. A checkbox for agreeing to the Terms and Privacy Policy must be selected before tapping the green "Create Account" button to complete the registration process.

Figure 4.9 shows the Home Dashboard featuring Create Meeting and Join buttons, meeting statistics, and quick action shortcuts including AI Chat, Summary, Translate, Schedule, History, and Profile for easy navigation.

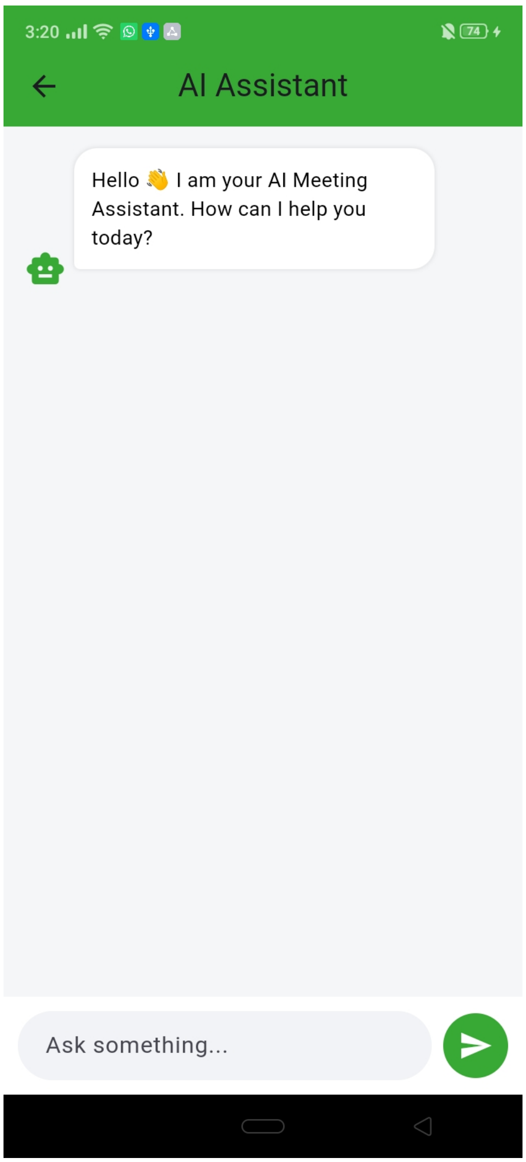

Figure 4.10 AI Assistant Screen

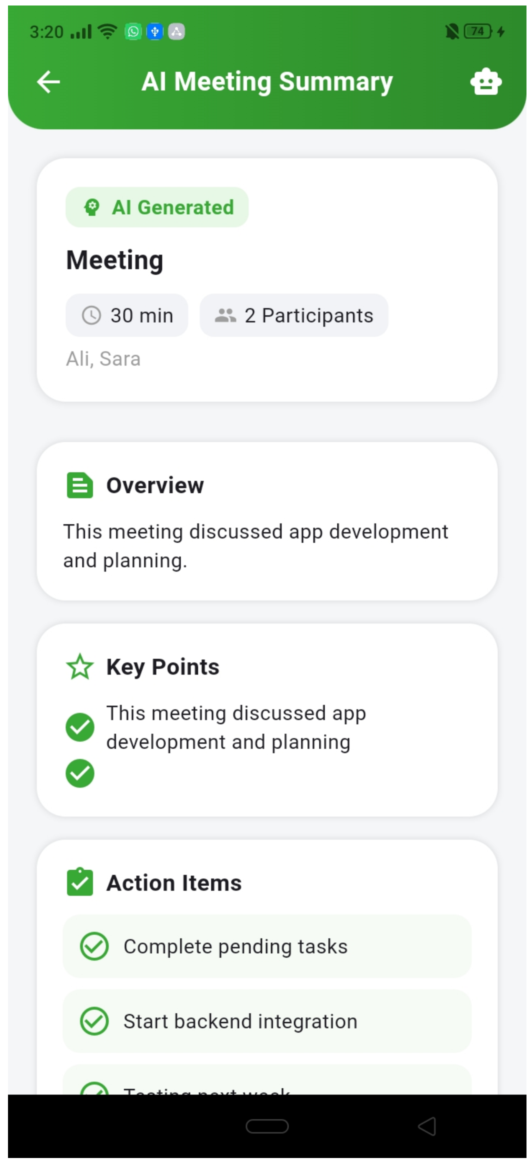

Figure 4.11 AI Meeting Summary Screen

Figure 4.10 shows the AI Assistant Screen with a chat interface where users can type natural language questions and receive answers retrieved directly from the meeting transcript.

Figure 4.11 Shows the auto-generated post-meeting report with meeting duration, participant count, and structured sections for Overview, Key Points, and Action Items.

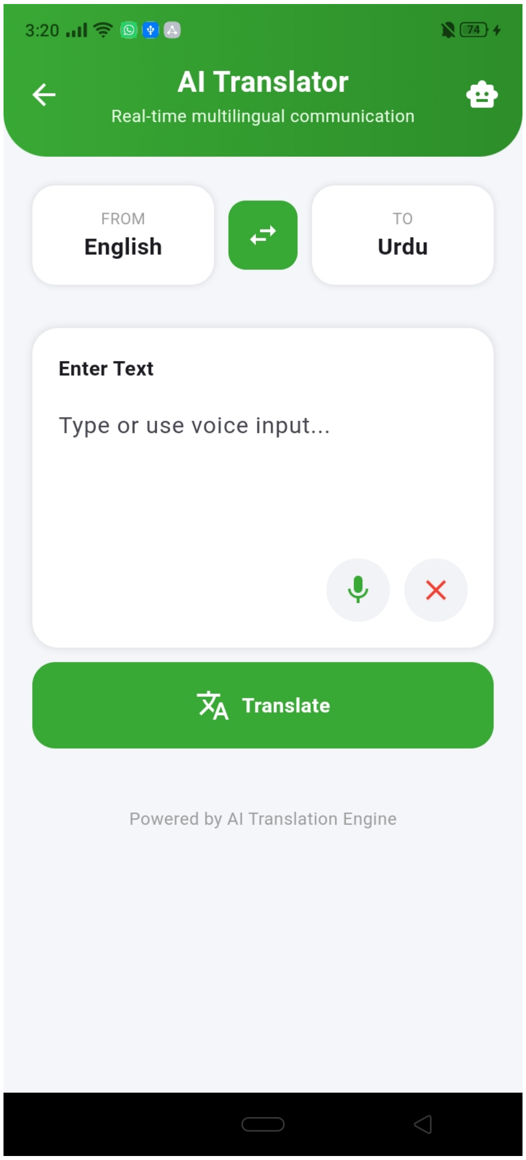

Figure 4.12 AI Translator Screen

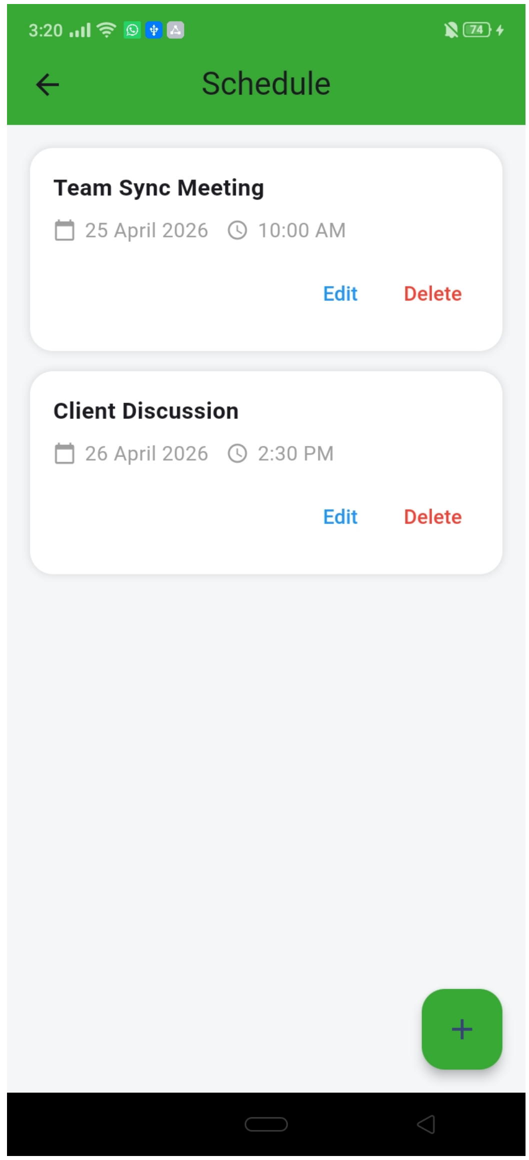

Figure 4.13 Schedule Screen

Figure 4.12 Shows the same AI Translator interface with English to Urdu language pair selected and voice input option available.

Figure 4.13 Shows upcoming scheduled meetings in cards displaying title, date, and time, with Edit and Delete options and a green ”+” button to add new meetings..

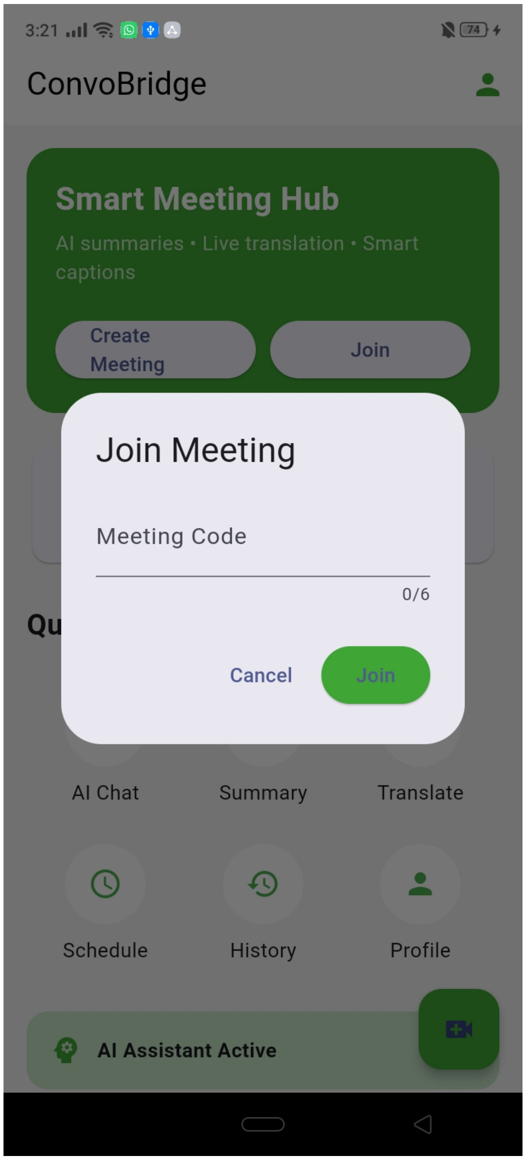

Figure 4.14 Join Meeting Dialog

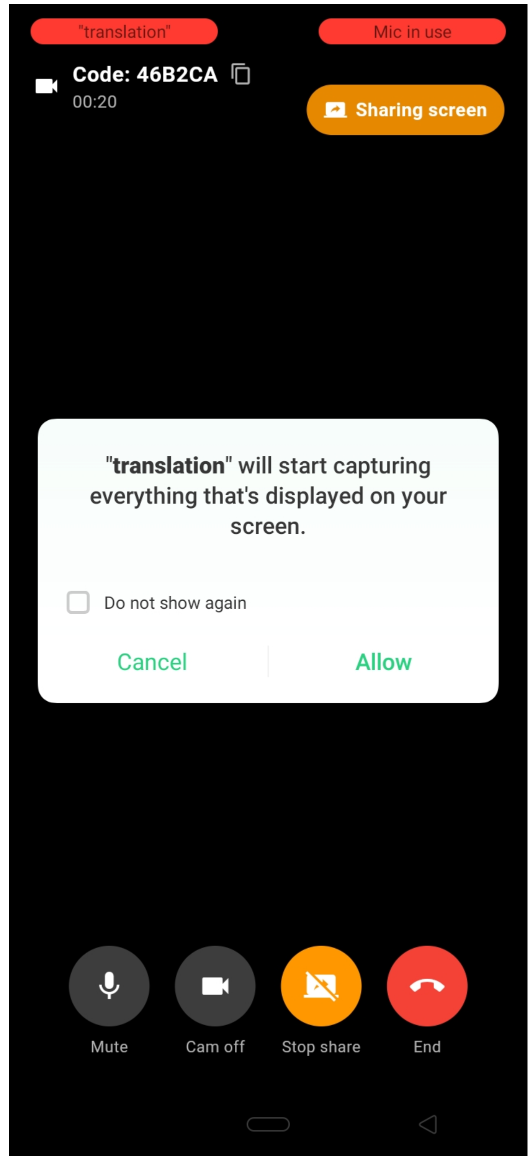

Figure 4.15 Create Account Screen

Figure 4.14Shows a popup over the Home Dashboard where the user enters a 6-character meeting code to join an active session, with Cancel and Join buttons.

Figure **??** Shows a system permission popup during an active meeting asking the user to allow screen capture. Controls at the bottom include Mute, Cam off, Stop share, and End.

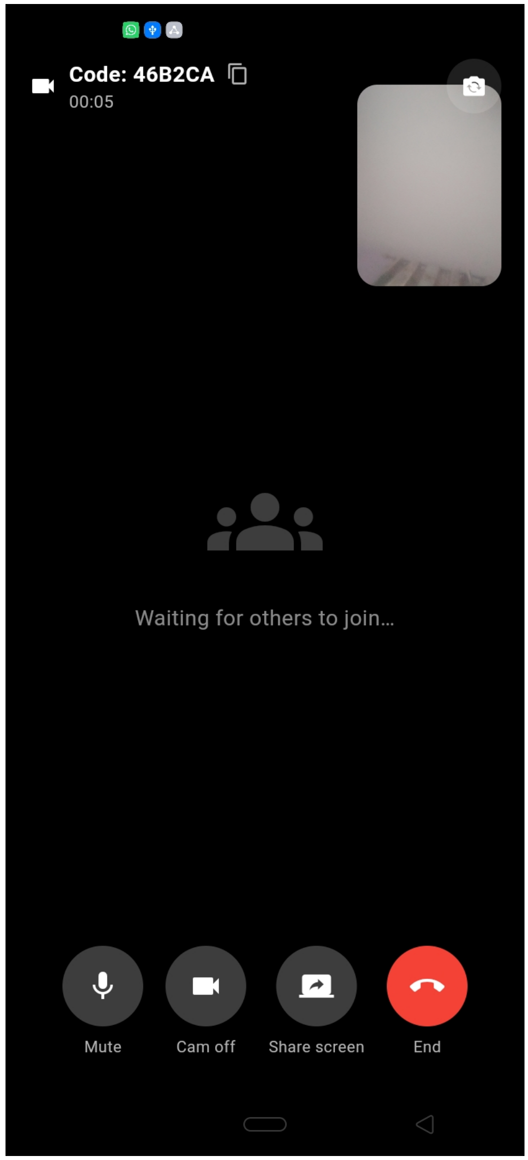

Figure 4.16 Active Meeting Screen

Figure 4.16Shows the live meeting session with the user's camera feed and a "Waiting for others to join" message. Bottom controls include Mute, Cam off, Share screen, and End.

4.5.4 Screen Objects and Actions

This section describes the major screen objects used in the ConvoBridge user interface and the actions associated with each object.

1: Buttons

The system includes buttons for navigation and interaction. The Login button authenticates the user and opens the dashboard. The Sign Up button directs users to registration. Create/Join Meeting buttons handle meeting access. The Back button returns to the previous screen, while the Logout button ends the session.

2: Text Input Fields

Text fields are used for user input. Email and Password fields are used for login. Full Name and Phone fields are used during registration. Meeting ID field is used to join meetings.

3: Icons

Icons provide quick navigation. The Profile icon opens settings, Chatbot icon opens AI assistant, Summary icon shows meeting summaries, History icon shows past meetings, and the Floating Action Button creates a new meeting.

4: Navigation Bar

A bottom navigation bar provides access to Dashboard, Meetings, History, and Profile screens for easy navigation.

5: Language Selection Chips

Language chips allow users to select translation languages such as English, Urdu, Arabic, and French. The selected language updates live captions instantly.

6: Live Captions Panel

Displays real-time speech-to-text output during meetings and updates continuously.

7: AI Listening Indicator

Shows that AI is actively processing speech and displays the number of active participants.

8: Status and Notifications

Displays system messages such as recording status, processing updates, and success/error alerts.

9: Security Notice

Shows that meetings are encrypted and securely managed.

10: Social Login Options

Allows users to sign in using Google, Facebook, or Apple accounts.

4.6 Algorithm

ConvoBridge integrates multiple intelligent algorithms across different modules to enable real-time speech processing, multilingual translation, AI summarization, and secure user management. These algorithms collectively ensure high performance, accuracy, and automation throughout the meeting workflow. The major algorithms used in this system are described below in step-wise form.

1. Real-Time Speech-to-Text Algorithm (Core Feature):

This algorithm runs on the Python backend and processes participant audio through the Whisper model in real time.

1. Capture raw audio stream from the participant's microphone via the Flutter application.
2. Pass the audio to the RNNoise module to suppress background noise and echo.
3. Send the cleaned audio to the Whisper Speech-to-Text engine.
4. Transcribe the spoken audio into text fragments with a latency of under two seconds.
5. Store each transcript fragment in the cloud database.
6. Return the transcribed text to the backend for translation processing.

2. Real-Time Translation Algorithm:

Steps for translating transcribed text into each participant's preferred language.

1. Receive the transcribed text fragment from the Speech-to-Text module.
2. Retrieve the preferred language of each active participant from the database.

3. Send the transcribed text to the Neural Machine Translation API.
4. Translate the text into each participant's individually selected target language.
5. Return the translated text to the Flutter application within three seconds.
6. Display the translated text as live multilingual subtitles on the active call screen.

3. Text-to-Speech Algorithm:

Steps for converting translated text into natural audio output.

1. Receive the translated text from the Translation module.
2. Identify the target language of the participant receiving the output.
3. Send the translated text to the Coqui TTS engine.
4. Synthesize natural human-like audio in the participant's preferred language.
5. Deliver the audio output to the participant within two seconds of translation.
6. Keep the TTS audio stream separate from the original speaker's audio stream.

4. AI Meeting Summarization Algorithm:

Steps for generating a structured post-meeting report using a GPT-based model.

1. Detect the meeting end event triggered by the host.
2. Fetch the complete meeting transcript from the cloud database.
3. Send the full transcript to the GPT-based summarization model.
4. Extract a concise meeting summary, key discussion highlights, and action items.
5. Structure the output into a formatted post-meeting report.
6. Store the report in the cloud database and make it available for in-app review and download within sixty seconds of the meeting ending.

5. Chatbot Query Processing Algorithm:

Steps for retrieving answers from the meeting transcript using natural language queries.

1. Receive the natural language query entered by the user through the chatbot panel.
2. Retrieve the current session transcript from the cloud database.
3. Pass the query and transcript to the GPT-based language model.
4. Identify and extract the most relevant answer from the transcript content.
5. Return the response to the user within three seconds of query submission.
6. Display the response in the chatbot panel without interrupting the ongoing session.

6. User Authentication Algorithm:

Steps for secure user registration and login using session token management.

1. Receive the registration or login request from the Flutter mobile application.
2. Validate the email format and check that the password meets minimum length requirements.
3. For registration, create a new user record and store the hashed password in the database.
4. For login, verify the entered credentials against the stored hashed password.
5. Upon successful verification, generate a session token and return it to the client.
6. Automatically invalidate the session token after 24 hours of inactivity.

4.7 Tools and Technologies

ConvoBridge is implemented using a modern, scalable tech stack combining mobile, backend, and AI technologies. The following tools and technologies were used during the design, development, and testing phases of the project.

Table 4.2 Tools and Technologies for ConvoBridge

S.No	Tool / Technology	Version	Purpose / Justification
1	Flutter	3.19	Cross-platform mobile application development for Android and iOS.
2	Dart	3.3	Programming language used with Flutter for frontend logic and UI.
3	Python	3.11	Backend runtime for AI module integration and REST API development.
4	FastAPI	0.110	Framework for building REST API endpoints connecting Flutter with AI modules.
5	Agora SDK	6.x	Real-time peer-to-peer audio and video call communication between participants.
6	Firebase Firestore	10.x	Generates unique meeting IDs and stores user profiles and transcripts.
7	Firebase Auth	10.x	Secure user authentication, login session management, and token generation.
8	Firebase Storage	10.x	Cloud storage for post-meeting summaries and files shared during sessions.
9	OpenAI Whisper	v3	Real-time Speech-to-Text transcription with multilingual support.
10	RNNoise	0.1.1	Real-time background noise suppression before audio reaches the STT module.
11	Coqui TTS	0.22	Natural multilingual Text-to-Speech audio synthesis in participant's preferred language.
12	Google Translate API	v2	Neural machine translation of transcribed text into each participant's language.
13	OpenAI GPT-4	GPT-4	Powers AI meeting summarization and chatbot assistant for transcript queries.
14	Socket.IO	4.7	Real-time signaling and event communication for meeting join, leave, and subtitles.
15	Google Cloud	Latest	Cloud hosting and deployment of Python backend and AI module APIs.
16	Figma	Latest	UI/UX design, screen prototyping, and mockup creation before development.
17	VS Code	1.88	Primary IDE for Flutter frontend and Python backend development.

Table 4.1 lists all the technologies used in ConvoBridge, covering mobile development, backend services, AI module integration, real-time communication, cloud storage, and deployment requirements.

4.8 External APIs/SDKs

This section describes the third-party APIs and SDKs used in the implementation of ConvoBridge. These external services are integrated into the system to support AI processing, real-time communication, translation, and cloud storage functionalities.

Table 4.3 External APIs/SDKs for ConvoBridge

S.No	API / SDK	Version	Purpose / Justification
1	Agora SDK	6.x	Provides real-time audio and video call functionality between meeting participants using peer-to-peer streaming.
2	Firebase Firestore SDK	10.x	Used to generate and store unique meeting IDs when a host creates a meeting, and to allow participants to join using that ID.
3	Firebase Auth SDK	10.x	Provides secure user authentication including email and password login and session token management.
4	Firebase Storage SDK	10.x	Used for uploading and retrieving shared files and post-meeting summary reports in the cloud.
5	OpenAI Whisper API	v3	Provides real-time Speech-to-Text transcription with support for multiple languages during active meeting sessions.
6	Google Translate API	v2	Translates transcribed meeting text into each participant's preferred language in real time with low latency.
7	Coqui TTS API	0.22	Converts translated text into natural human-like audio output in the participant's selected language.
8	OpenAI GPT-4 API	GPT-4	Powers the AI meeting summarization module to generate summaries, key highlights, and action items from the meeting transcript.
9	OpenAI GPT-4 API	GPT-4	Also used to power the Chatbot Assistant that retrieves answers from the meeting transcript based on natural language queries.
10	Socket.IO	4.7	Provides real-time WebSocket-based communication for meeting signaling events including join, leave, mute, and subtitle updates.
11	RNNoise Library	0.1.1	An open-source noise suppression library used to remove background noise from participant audio before STT processing.

Table 4.2 lists all the external APIs and SDKs integrated into ConvoBridge. Each service was selected based on its performance, multilingual support, ease of integration with Flutter and Python, and suitability for real-time AI-powered communication.

4.9 Deployment

ConvoBridge is deployed using a cloud-based architecture that ensures scalability, reliability, and accessibility for all users. The system is divided into two main deployment environments: the Flutter mobile application deployed on Android and iOS platforms, and the Python backend deployed on Google Cloud.

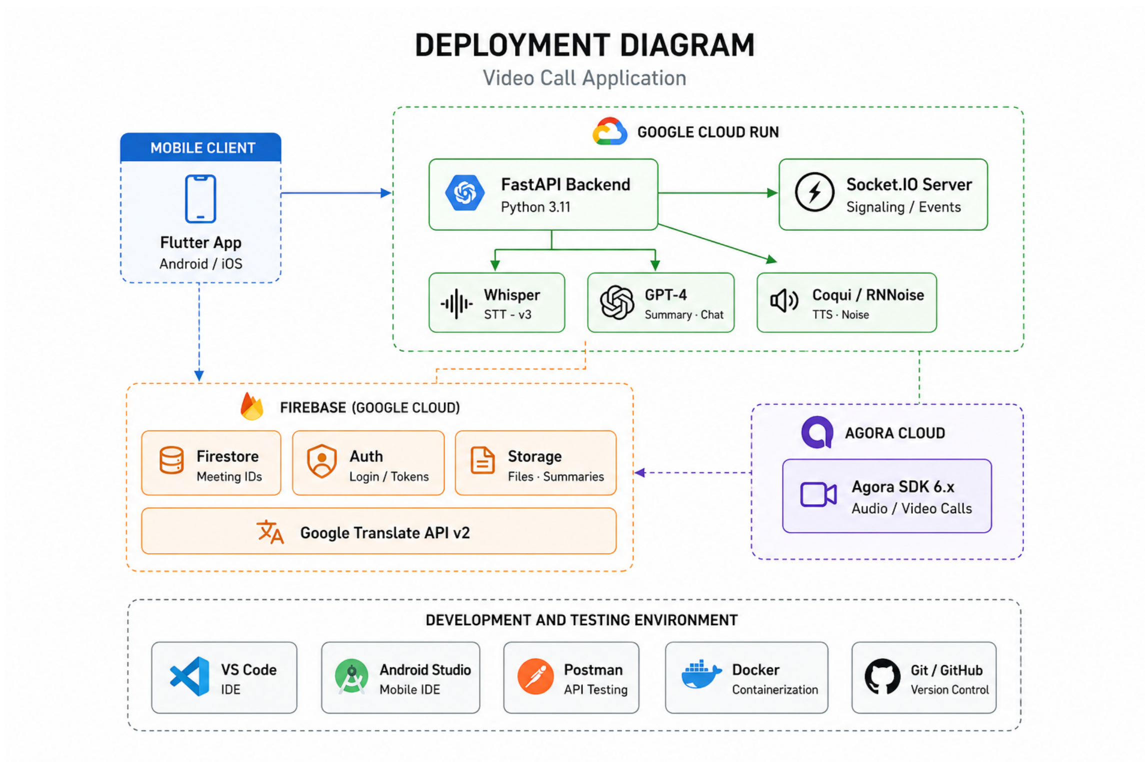

Figure 4.17 Deployment Environment for ConvoBridge

Table 4.4 Deployment Environment for ConvoBridge

S.No	Component	Platform	Details
1	Flutter Mobile App	Android / iOS	Built using Flutter 3.19 and deployed as an APK for Android and IPA for iOS. Tested on Android 10 and above and iOS 14 and above.
2	Python Backend	Google Cloud Run	The FastAPI backend is containerized using Docker and deployed on Google Cloud Run for scalable and serverless hosting.
3	AI Modules	Google Cloud	Whisper STT, RNNoise, Coqui TTS, and GPT-4 API calls are handled through the Python backend hosted on Google Cloud.
4	Firebase Firestore	Firebase Console	Used for real-time cloud database storing meeting IDs, user profiles, and transcripts. Managed through the Firebase Console.
5	Firebase Auth	Firebase Console	Handles user authentication and session management deployed through Firebase Authentication service.
6	Firebase Storage	Firebase Console	Stores post-meeting summaries and shared files. Deployed through Firebase Storage with cloud access rules.
7	Agora SDK	Agora Cloud	Real-time audio and video communication is handled through Agora's cloud infrastructure with automatic scaling.
8	Socket.IO Server	Google Cloud	Socket.IO signaling server is deployed alongside the FastAPI backend on Google Cloud Run for real-time event communication.

Development and Testing Environment:

Table 4.5 Development and Testing Environment

S.No	Tool	Version	Purpose
1	Android Studio	Hedgehog	Used for Flutter development, Android emulation, and APK build generation.
2	VS Code	1.88	Primary code editor for Flutter frontend and Python backend development.
3	Postman	10.x	Used for testing and validating all REST API endpoints during development.
4	Docker	Latest	Used to containerize the Python backend for consistent deployment on Google Cloud.
5	GitHub	Latest	Used for version control, collaborative development, and CI/CD pipeline management.
6	Firebase Console	Latest	Used for monitoring database, authentication, and storage services during live testing.

4.10 Chapter Summary

This chapter presented the complete design and architecture of the ConvoBridge system. The system modules were organized into two groups: the Client Mobile Application built in Flutter and the Backend Application built in Python with FastAPI. The architectural design follows a three-layered Client-Server structure consisting of the Presentation Layer, Application Layer, and Data Layer. Design models including the Activity Diagram, Use Case Diagram, Sequence Diagram, Class Diagram, and State Transition Diagram were presented to illustrate system behavior, interactions, and data flow. The data design described all major entities and their attributes through a structured data dictionary. The human interface design explained the user experience across all major screens. Finally, the tools and technologies, external APIs, and deployment environments used in the implementation of ConvoBridge were documented.

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